

1.1.1.1. Chart Areas

The Chart Area of each panel (**Main** and **Dual**) is designed for the following purposes:

- to display electronic charts showing classes of chart information (determined on the **Charts** panel);
- to display the ownship symbol with motion vectors;
- to display the route loaded for proceeding by in the Navigation Mode;
- to display the AIS objects (only the **Main** panel).

Displayed in the top left corner of the Chart Area is the panel orientation symbol  : an arrow showing direction to the north.

The bottom right corner of the electronic chart displays an angle shaped indicator .

The line is a graphic presentation of the maximum possible error of plotting objects on the chart. When a chart is displayed on the scale of the original, the linear size of the error is taken to be 2 mm. As the scale is growing, the linear dimensions of the indicator increase showing to which extent the chart information can be relied on.

1.1.1.2. Toolbars

The **Main** chart panel toolbar contains the following control buttons:



- to return to the display of the ownship symbol;



- to set the ship symbol in the position of the cursor which appears when this mode is turned on;



- to select chart display orientation (North Up/Head Up/Course Up);



- to select the ship symbol motion mode (True/Relative);



- to increase the chart display scale by a fixed scale value;



- to decrease the chart display scale by a fixed scale value;



- to display the chart on the original scale;



- to call the Info cursor for obtaining information;



- to display HCRF format chart viewing panel;



- to turn on/off Quick Distance Tool and select the mode of distance measurement;



- to turn on the display of the chart over the radar picture;



- Turn off Radar, AIS, ARPA overlay (added in ECDIS, in RADAR exist)



- to open the **Integrator** toolbar;



- to select the screen colour palette to suit the outside illumination;



- to turn on/off virtual keyboard.



- to load (create/edit) Default Configuration



- to load (create/edit) User Configurations;



- to enable/disable display CATZOC Overlay



- to turn on/off the MOB mode.

The **Dual** chart panel toolbar contains the following control buttons:



- to return to the display of the ownship symbol;



- to set the ship symbol in the position of the cursor which appears when this mode is turned on;



- to select chart display orientation (North Up/Head Up/Course Up);



- to select the ship symbol motion mode (True/Relative);



- to increase the chart display scale by a fixed scale value;



- to decrease the chart display scale by a fixed scale value;



- to display the chart on the original scale;



- to turn on the Auto Zoom function;



- to arrange Chart panels horizontally;



- to arrange Chart panels vertically;



- to turn on the Docking Mode;



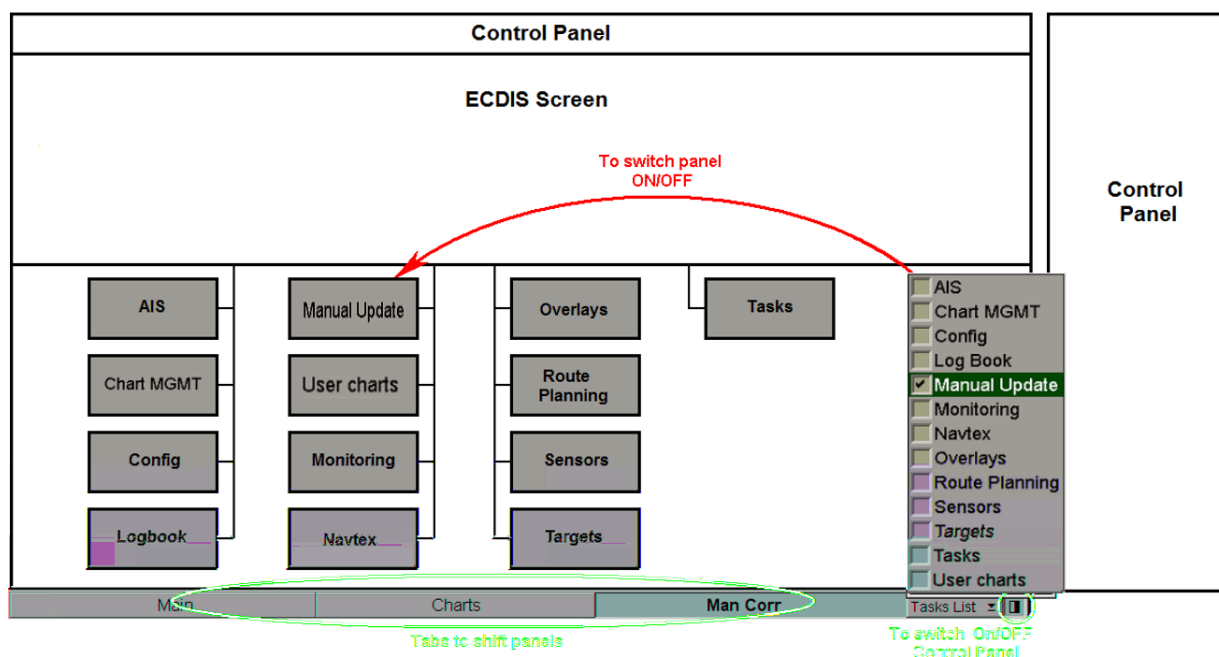
- to turn on the history of the Docking Mode.

1.2. User Interface Components

1.2.1. Panels

There is direct access to panels from the task. For the ECDIS task, this is primarily the Control panel, which shows operational information and contains the most important task controls.

In the ECDIS task, apart from the Control panel, there are 13 functional panels, each designed for its special purpose within the ECDIS task:



Panels (except the Control panel) are switched with tabs in the bottom part of the screen.

For the operator's convenience, the system implements a function for saving the current condition of the Main chart panel when the functional panels are switched. This function's operation can be shown in a table:

PANEL NAME	TO OPEN PANEL	TO SWITCH THE OPEN PANEL WITH THE TAB
Main	A	S
Dual	S	H
AIS	A	H
Chart MGMT	P	K
Config	P	P
Log Book	P	K
Manual Update	L	K
User Charts	L	K
Monitoring	P	P
Navtex	L	K
Overlays	P	K
Route Planning	L*	K

1.3.10. Primary Window

Prim	50° 00.212 N
GPS 1	001° 00.459 E

The Primary window is intended for the display of the primary positioning system and coordinates obtained from it. For the principles of displaying primary positioning data, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors, paragraph Display of Navigational (Essential) Information*.

Prim	50° 00.000 N
GPS 1	000° 57.472 W

GARMIN

The position source is specified in the left bottom part of the window. As a free cursor is positioned on the position source name, the source alias entered in the System Configuration utility is displayed in the pop-up prompt window.

Prim	50° 00.212 N
DGPS 1	001° 00.459 E

With the GPS operating in the differential mode, the prefix "D" is added before its name.

Prim	60° 01.595 N
GPS 1	029° 16.496 E

Offset:
00° 00.396 S
000° 00.273 W

If there is an offset to the ship position coordinates, the window displays corrected coordinates and a special symbol: the yellow triangle. As a free cursor is positioned on this triangle, the correction value is displayed in the pop-up prompt window.

If the RMC (IEC 61162 ed.4) message is available, the MFD processes it and displays navigational status in the **Primary** window. The colour in which the coordinates value is shown indicates the position quality:

- Black – safe;
- Orange – caution;
- Red – unsafe (this event is recorded in the Log Book).

SAFE	CAUTION	UNSAFE
Prim 27° 22.505 N Position state - safe	Prim 27° 22.505 N DGPS Position state - caution	Prim 49° 37.169 N GPS 1 002° 57.167 W Sec: No Position state - unsafe

1.3.11. Secondary Window

Sec: GPS 2	167.5° 0 m
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The **Secondary** window is intended for the display of the secondary positioning system, and the bearing and range to the position obtained for the Primary positioning system. For the principles of displaying secondary positioning system data, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors, paragraph Display of Navigational (Essential) Information*.

Sec: GPS 2	90.0° - 178 m
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TRIMBLE

In the list which will open up, select the necessary menu line to set the screen colour palette.

ATTENTION! Use of the brightness control may inhibit visibility of information at night.

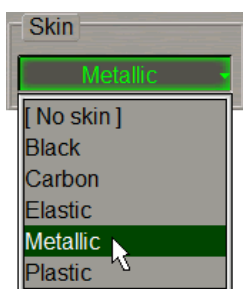
There is another way to switch palettes. Press successively <DAY/NT> on the ES6 keyboard. Palettes will be switched in the order of their arrangement in the **Palette** group of the **General** page.

3.5.2. Setting of Screen Graphic Presentation

The **Skin** group is designed for selecting the screen graphic presentation:

- No skin;
- **Black;**
- **Carbon;**
- **Elastic;**
- **Metallic;**
- **Plastic.**

In the **Skin** group, press the button with the name of the set screen graphic presentation.



3.5.3. Switching Graphic Cursor Type

The graphics cursor is used during the operation of some of ECDIS task functions and represents an intersection of lines corresponding to the latitude and longitude of the given point, but the form of the graphics cursor can be changed as required. The **Chart panel cursor** group serves for setting the following types of the graphics cursor:

- Short – the cursor is shown in the form of a small cross;
- Long – the cursor is shown in the form of an intersection of two lines across the entire Chart pa



In the **Chart panel cursor** group, select the appropriate checkbox to select the graphic cursor type.

3.6.4. Setting Ship Symbol Motion Modes

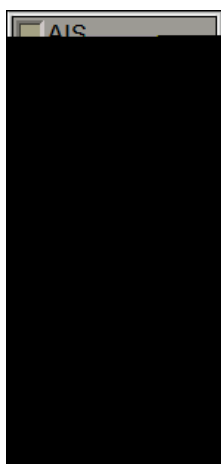
To select the ship symbol motion mode, press the button in the toolbar of the Chart panel:

-    – to select relative motion mode;
-    – to select true motion mode.

There is another way to select the ship symbol motion mode. Press successively the <TM/RM> key on ES6 keyboard. The ship symbol motion mode will be changed to the reverse.

3.6.5. Setting Ship Display and Its Motion Parameters on ECDIS Task Screen

Open the **Monitoring** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Monitoring** panel, which will open up, to switch to the **Route Monitoring** page:



To turn on/off the display of own ship and its motion parameters, press the following buttons in the **Ship** group:

- **Headline** – to display the own ship compass heading line;
- **COG vector** – to display the motion vector (over the ground);
- **HDG vector** – to display the vector of motion relative to the water (log speed and compass heading).

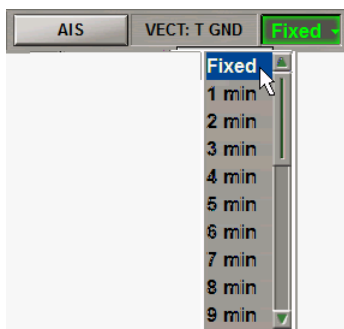


Select from the drop-down lists the following settings:

- **Ship by ...** – to display the own ship (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Ship Motion Data, paragraph Ship Symbol on the Electronic Charts*) on the Chart panel as:
 - **symbol** – the ship symbol (regardless of the scale);
 - **contour** – the ship contour (when the current screen scale is commensurate with the ship dimensions).
- **Align by ...** – to switch orientation (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Ship Motion Data, paragraph Own Ship Motion Vectors*) of the own ship symbol display along:
 - **HDG** – the vector of the ship's motion as per the heading detector (gyro) reading;
 - **COG** – the ship's true motion (COG) vector.
- **Wind vector** – to select true wind vector display mode (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Other Sensors, paragraph Wind Interface*).

3.6.6. Setting of Lengths of Own Ship and Target Vectors

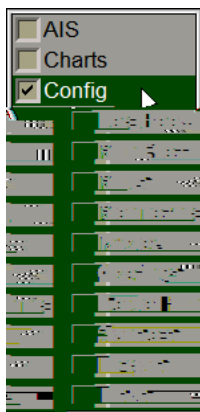
Press vectors button in the **Vectors** window on the Control panel (the button shows the current vector length).



Select the required vector length (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 3, section Own Ship Motion Data, paragraph Own Ship Motion Vectors*) from the list which will open up and press the left trackball button.

3.7. Setting Time Zone

Open the **Config** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Config** panel which will open up, to switch to the **Time Zone** page.

The screenshot shows the 'Time Zone' tab in the configuration panel. It is divided into two main sections: 'Current time zone settings' and 'Scheduled time zone change'.
 In the 'Current time zone settings' section, there are three input fields: 'UTC time:' showing '10-12-15 16:00', 'Ship's time:' showing '10-12-15 16:00', and 'Time zone:' showing '00:00'.
 In the 'Scheduled time zone change' section, there is a radio button labeled 'Change at ship's time:' which is selected. Below it are three input fields: 'Change at ship's time:' showing '10-12-15 16:00', 'New ship's time will be:' showing '10-12-15 16:00', and 'New time zone will be:' showing '00:00'.

3.7.1. Setting Current Time Zone

The **Current time zone settings** group contains the following elements:

- **UTC time** – line for the display of the current UTC time;
- **Ship's time** – line for the display of the current ship time;
- **Time zone** – line for the display of the time zone (difference between the current ship and UTC time).

This screenshot shows the 'Current time zone settings' panel. It contains three input fields: 'UTC time:' with the value '30-08-12 10:31', 'Ship's time:' with the value '30-08-12 12:31', and 'Time zone:' with the value '02:00 E'.

Use the **Time zone** line to enter the ship's time zone (from 14E to 12W). Press <ENTER>: the time zone value will be set, whereas the ship time value in the **Ship's time** line will change to suit the entered time zone. As the ship time is entered in the **Ship's time** line, the time zone will change in the **Time zone** line.

3.7.2. Setting Time Zone Change

The **Schedule time zone change** group contains the checkbox for enabling (checkbox selected) the change of time zone function and a line for changing settings:

- **Change at ship's time** – line for the input and display of the ship time when the time zone is required to be changed (clock put forward/back);
- **New ship's time will be** – line for the input and display of the new ship time;
- **New time zone will be** – line for the input and display of the new time zone.

This screenshot shows the 'Scheduled time zone change' panel. It has three input fields: 'Change at ship's time:' with the value '30-08-12 10:27', 'New ship's time will be:' with the value '30-08-12 10:27', and 'New time zone will be:' with the value '00:00'. There is also a radio button labeled 'Change at ship's time:' which is selected.

Use the **New time zone will be** line of the **Schedule time zone change** group to set the new time zone value. Press <ENTER>. Use the **Change at ship time** line to set the ship time value when the time zone is expected to change.

This screenshot shows the 'Scheduled time zone change' panel with the 'Change at ship's time:' radio button selected. The input fields show: 'Change at ship's time:' with the value '30-08-12 12:00', 'New ship's time will be:' with the value '30-08-12 15:00', and 'New time zone will be:' with the value '03:00 E'.

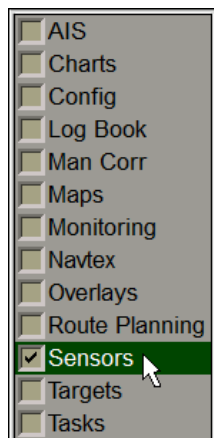
Press the <ENTER> keyboard key. Select the activated **Change at ship time** checkbox. At the set time, the time zone will be changed to suit the value entered for it.

4. Navigation Tasks

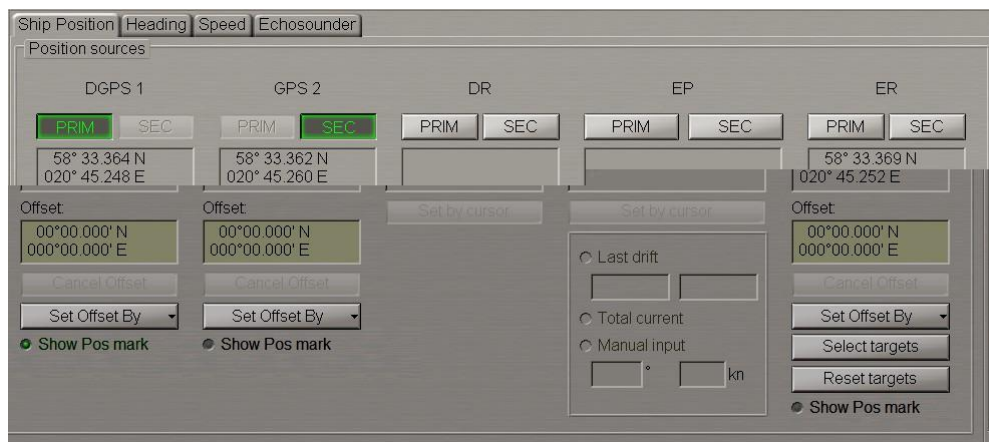
4.1. Setting Position Sources

ATTENTION! Setting of position sources is available only on the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel which will open up, to switch to the **Ship Position** page:



The **Ship Position** page is designed for selecting the position source. The **Ship position** page displays all the connected coordinates sensors, as well as the Dead Reckoning (DR) and Echo Reference (ER) modes.

4.1.1. Setting Electronic Position Fixing System (EPFS)

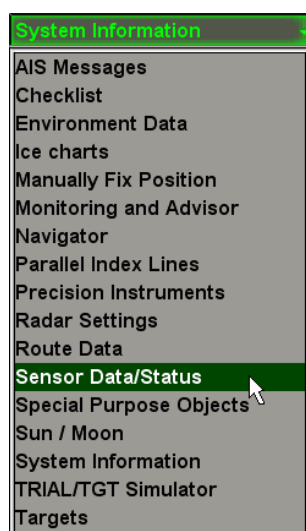


The group of each EPFS source contains the following elements:

- **PRIM** – to select the source as the primary positioning system; the window is intended for displaying the current coordinates received from the positioning system;
- **SEC** – to select the source as the secondary positioning system; the window is intended for displaying the current coordinates received from the positioning system;
- Position window for the coordinates display;
- **Offset** – window for the input and display of the ship position coordinates offset;
- **Cancel Offset** – to cancel the entered corrections;
- **Set Offset By** – to enter the correction by specifying new coordinates with the ECDIS task graphics cursor.

Press the **PRIM** and **SEC** buttons to select the primary and secondary positioning systems.

To monitor parameters of the EPFS used for the primary positioning system, press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Sensor Data/Status** line and press the left trackball button.

Use the tab in the top part of the **Sensor Data/Status** display panel which will open up, to switch to the **Primary Status** page.

Sensor data/status	
Primary Status AIS VDL Data	
Fixed UTC	21 : 36 : 24
Latitude	56° 48.766 N
Longitude	018° 21.135 E
Quality	DGPS SPS
Satellites	5
HDOP	1.0
Data age	10.0
Station ID	0000
RMS 95%	20 m

The **Primary Status** page is used for presenting the following data from the EPFS used as the primary positioning system:

- **Fixed UTC** – UTC time which the data is provided for;
- **Latitude** – latitude coordinate;
- **Longitude** – longitude coordinate;
- **Quality** – positioning quality indicator (GPS operating mode);
- **Satellites** – number of satellites used for the positioning;
- **HDOP** – HDOP (Horizontal Dilution of Precision) value;
- **Data age** – age of differential corrections;
- **Station ID** – name of the station which transmits differential corrections for the DGPS used by the primary positioning system (where the differential mode is used);
- **RMS 95%** – ship position 95% RMS error (for the 95% RMS error calculation algorithm, see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors*, paragraph *Positioning*, under *RMS Calculations for Different Positioning Systems*).

In the absence of data from the PS positioning systems (or in the absence of a valid GGA sentence), there are empty fields on the display.

To enter the EPFS offset, use the following procedure:

1. For manual input. Position the cursor in the **Offset** window on the minute value and press the left trackball button to activate the window. Enter the coordinate offset and press <ENTER>.

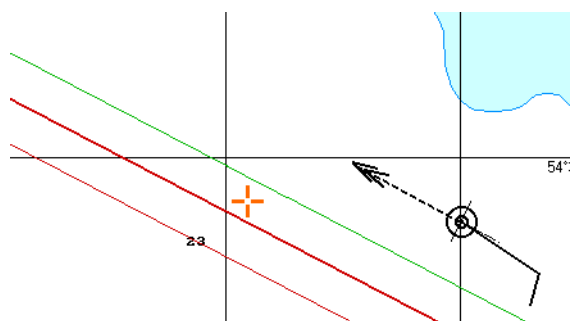
The screenshot shows the 'GPS 1' status window. At the top, there are two buttons: 'PRIM' (highlighted in green) and 'SEC'. Below them, the main position is displayed as '50° 11.478 N' and '000° 06.535 E'. Underneath is the 'Offset' section, which shows '00° 10.000' N' and '000° 00.000' E'. The '10' in the latitude offset is highlighted in green. Below the offset fields are two buttons: 'Cancel Offset' and 'Set Offset By' with a dropdown arrow.

The ship symbol will move to the corrected coordinates, the coordinate offset will be displayed in the **Offset** window, whilst the position window will show corrected coordinates from the position source.

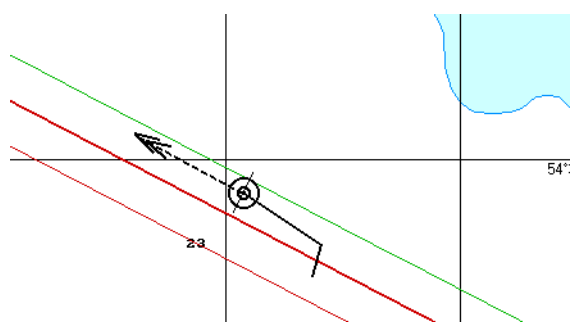
2. Input of the EPFS offset by the cursor. Press the **Set Offset By** button in the position source group and select **Cursor**.

This screenshot shows the same 'GPS 1' status window as before, but with changes. The main position is now '55° 11.684 N' and '000° 08.576 E'. The 'Offset' section still shows '00° 00.000' N' and '000° 00.000' E'. The 'Set Offset By' button is now highlighted in green, and its dropdown menu is open, showing 'Cursor' as the selected option. A mouse cursor is pointing at the 'Cursor' option.

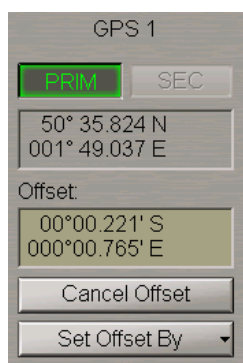
Move the graphics cursor which will appear, to the corrected ship position coordinates.



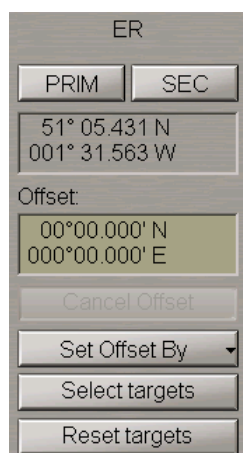
Press the left trackball button. The ship symbol will move to the specified coordinates.



The coordinate offset will be shown in the **Offset** window, whilst the position window will display corrected coordinates from the position systems.



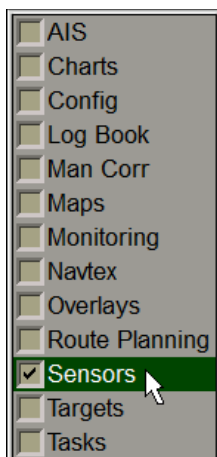
4.1.2. Setting ER Mode



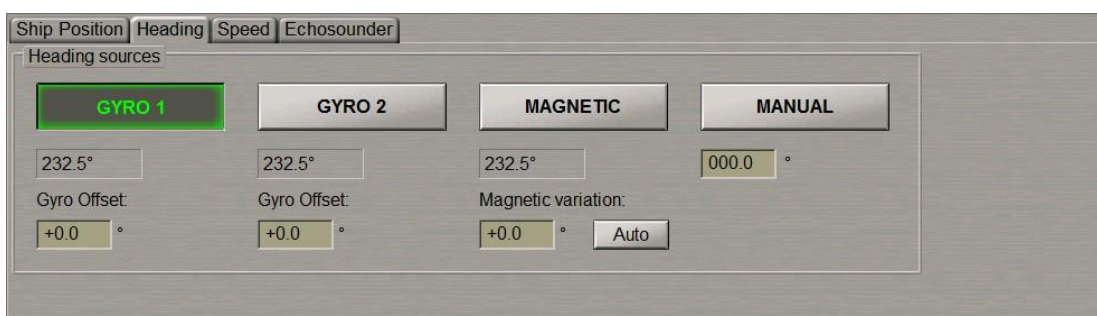
4.2. Setting Heading Source

ATTENTION! Setting of heading sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Heading** page:



The **Heading** page is designed for selection of a heading source. The **Heading** page shows all the connected heading sensors.

The group of each heading source contains the following elements:

- **GYRO 1** – to select this sensor as the heading source; the window is intended for displaying a heading value from the sensor;
- **Gyro Offset** – for the manual input of the gyro offset (for gyro compass);
- **Magnetic variation** – for the input of the magnetic variation from the database (the Auto button), or manually in the input window (for the magnetic compass);
- **MANUAL** – to select the use of manual heading input; the window is intended for the heading input.

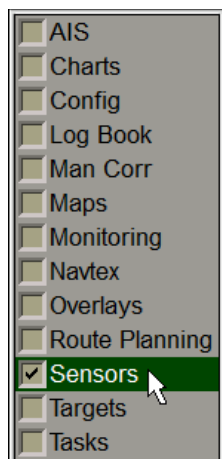
Press the button with the source name to select the main source of heading whose data will henceforth be used for making all the calculations in the system.

If necessary, enter errors (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 2, section Navigational Sensors, paragraph Compensations for Navigational Data*) for gyro and variation for magnetic compass.

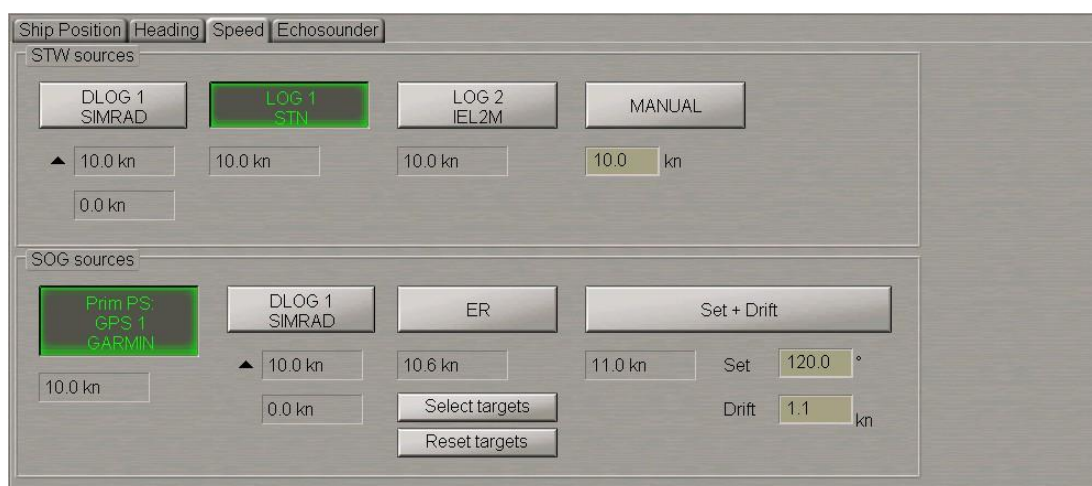
4.3. Setting Speed Sources

ATTENTION! Setting of speed sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Speed** page.



The **Speed** page is designed for selecting a speed source.

4.3.1. Setting Speed-Through-Water Source



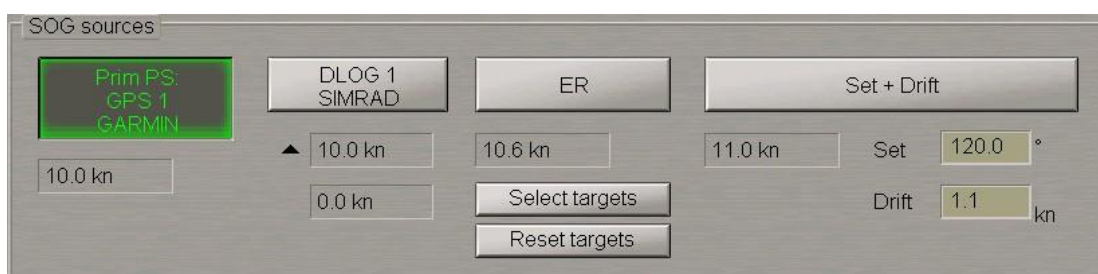
The **STW sources** group is designed for selecting a source of speed through the water. The group shows all the connected speed-through-water sensors.

The group of each speed-through-water source can contain the following elements:

- **DLOG** – to select the DLOG as a source of STW; the boxes are intended for displaying longitudinal and transverse components of the speed-through-water vector, the arrows showing the direction of components;
- **LOG** – to select the LOG as a source of speed through water; the window is intended for displaying STW value from the log;
- **MANUAL** – to select the use of manual STW input; the window is intended for the speed input.

Press the button with the source name to select the source of speed through the water, whose data will henceforth be used for making all the calculations in the system.

4.3.2. Setting Speed-Over-Ground Source



The **SOG sources** group is designed for selecting a source of speed over ground. The group shows all the connected speed-over-ground sensors.

The group of each speed-over-ground source can contain the following elements:

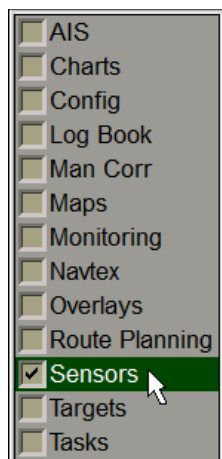
- **Prim PS: GPS 1** – to select the EPFS as a source of SOG; the window is intended for displaying SOG value from the EPFS;
- **DLOG** – to select the DLOG as a source of SOG; the boxes are intended for displaying longitudinal and transverse components of the SOG vector, the arrows showing the direction of components;
- **ER** – to select the ER mode as a source of SOG; the window is intended for displaying the SOG value; the buttons are intended for:
 - **Select targets** – to select targets as fixed reference points (the button is enabled if at least one steadily tracked target is available);
 - **Reset targets** – to cancel reference points.
- **STW + Drift** – to select STW as SOG, manual drift input taken into account. The window is intended for displaying SOG value. The input boxes are intended for:
 - **Set** – to display and manually enter the set;
 - **Drift** – to display and manually enter the drift.

Press the button with the source name to select the source of speed over ground whose data will henceforth be used for making all the calculations in the system.

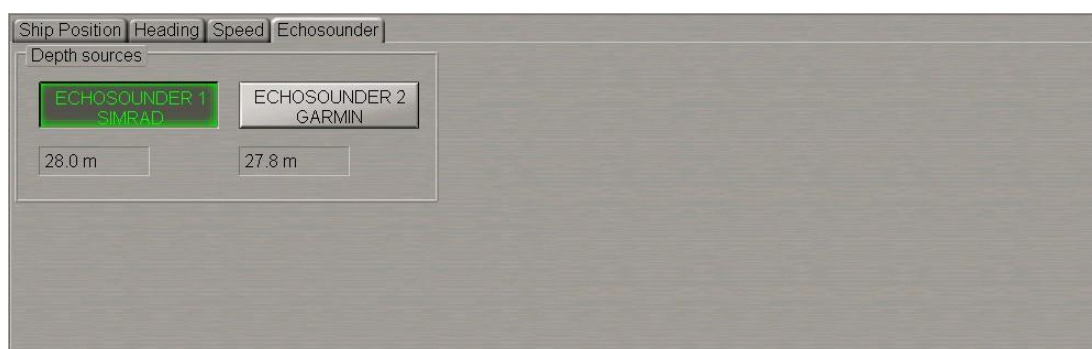
4.4. Setting Depth Source

ATTENTION! Setting of depth sources is available only at the station with the MASTER status.

Open the **Sensors** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Sensors** panel to switch to the **Echosounder** page.



The **Echosounder** page is designed for selecting a depth source. The **Echosounder** page shows all the connected depth sensors.

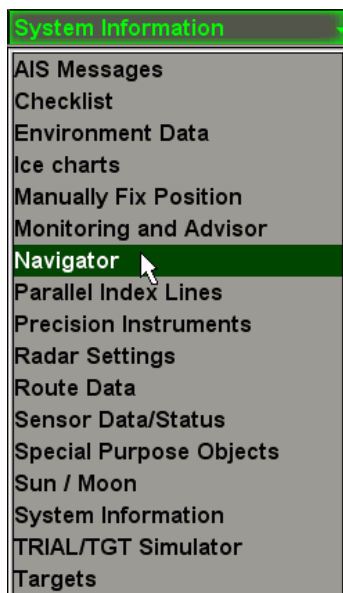
The group of each depth source can contain the following elements:

- **ECHOSOUNDER 1** – to select this sensor as a depth source; the window is intended for displaying the depth value from the sensor.

Press the button with the source name to select the main source of depth whose data will henceforth be used for making all the calculations in the system.

4.5. Obtaining Current Navigational Data Information

Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Navigator** line and press the left trackball button.

Navigator	
HDG ▾	269.9°
STW ▾	5.0 kn
COG ▾	269.9°
SOG ▾	5.0 kn
ROT ▾	1.0/min PORT
XTD ▾	135 m STBD

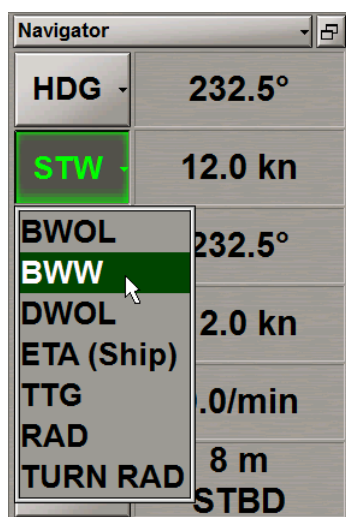
The **Navigator** display is intended for enlarged presentation of the following selectable data:

- HDG (Heading);
- STW;
- COG;
- SOG;
- ROT;
- XTD;
- BWOL (Bearing to Wheel Over Line);
- BWW (Bearing Waypoint to Waypoint);
- DWOL (Distance to Wheel Over Line);
- ETA;

- TTG (Time To Go);
- Radius (turn radius at the next waypoint);
- Turn Radius;
- DPT.

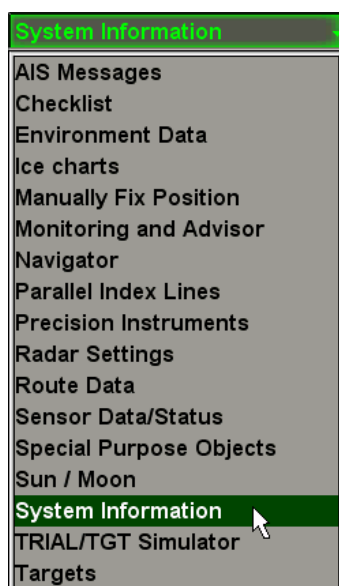
Only six values can be displayed simultaneously on the display.

Press the button in the left-hand part of the display and from the pull-down list, select navigational data which is required to be shown on the display.



Make up the **Navigator** display configuration which suits you best.

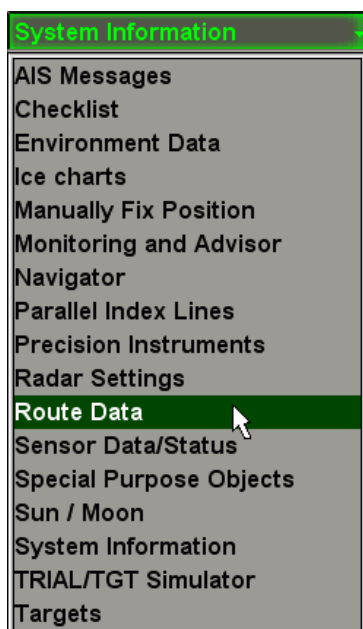
To display depth from the echosounder on the ECDIS task screen, press the button with the name of the set display in the **Display Panel** window of the Control panel.



4.6.5. Obtaining Current Information on Monitored Route and Schedule

4.6.5.1. Route Data

Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Route Data** line and press the left trackball button.

Use the tab in the top part of the **Route Data** display which will open up, to switch to the **Route Data** page.

Route Data	
Route Data	Schedule RDV
Route	NONAME04
To WPT 1	
BWW	270.0°
XTD	STBD - 107 m
XTL	PORT 185 m
	STBD 185 m
BWOL	265.8°
DWOL	0.79 NM
ETA (UTC)	19-08-25 09:26
TTG	23 min 42 s
Next WPT 2	
Next BWW	290.6°
RAD	0.10 NM

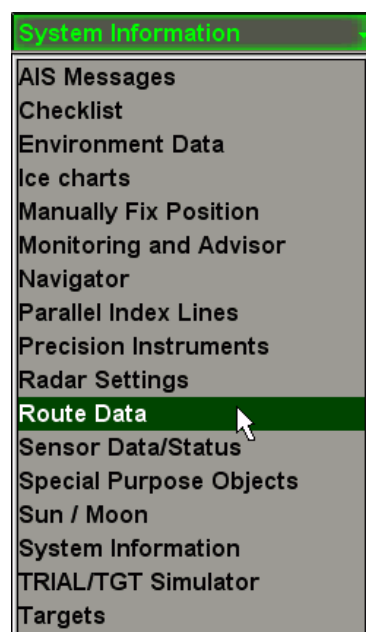
The **Route Data** page is intended for the display of data as the ship is proceeding along the route to the next waypoint (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 5, section Route Monitoring, paragraph Sailing Along the Route*).

The page contains the following data:

- **Route** – name of the monitored route; as the End of route warning is generated, the word “Route” is shown in the orange colour;
- **To WPT** – number and name of the current WPT which the ship is proceeding to; the WPT name is shown in black if the WPT is selected automatically, and in the blue colour if selected manually;
- **BWW** – direction of the monitored route leg which the ship is proceeding by if the leg is of the RL type; or the current course in the case of the GC leg;
- **XTL** - Cross Track Limits from the current leg;
- **XTD** – actual distance from the current route leg; as the **Off track** alarm is generated, the letters “XTD” are shown in red;
- **BWOL** – bearing to the current WOL;
- **DWOL** – distance to the current WOL;
- **ETA** – the estimated time (UTC/Ship depending on the setting made in the Control panel) of arrival in the current WPT, the remaining distance and current speed (SOG) taken into account;
- **TTG** – time to go to the current waypoint calculated with regard to the actual speed (SOG);
- **Next WPT** – number of the next WPT;
- **BWW** – direction of the next route leg;
- **RAD** – arc radius on the turn to the next route leg.

4.6.5.2. Schedule Information

Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Route Data** line and press the left trackball button.

Use the tab in the top part of the **Route Data** panel which will open up, to switch to the **Schedule** page.

Route Data	
Route Data Schedule RDV	
To WPT:	20
Schedule Current SOG	
ETA (UTC)	09-12-2015 05:15:32
TTG	0 d 12 h 33 min
DWOL	168.1 NM
PTA (00:00)	09-12-2015 05:45:27
Quick calculation	
STG (kn)	PTA (Ship time)
8.0	12-12-15 00:28
12.0	09-12-15 12:21

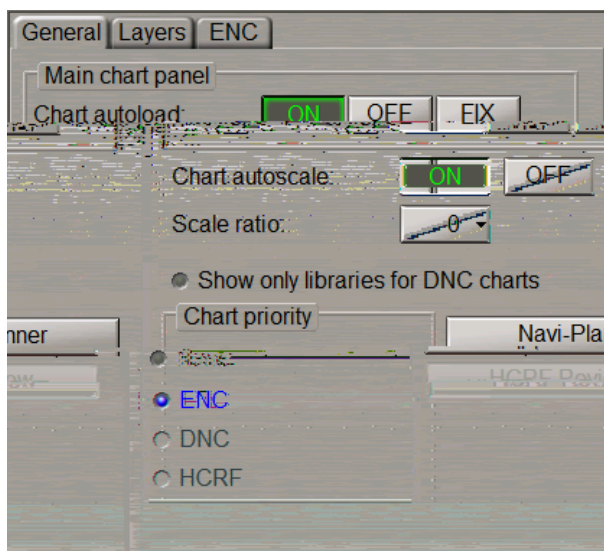
The **Schedule** page serves for presenting information related to sailing according to the selected schedule (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 5, section Route Monitoring*, paragraph *Sailing Along the Route*). For the display of data, it is necessary to load a route on the **Route Monitoring** page of the **Monitoring** panel (see above *Loading Route*). The schedule elements whose values differ from those set on the **Route Monitoring** page of the **Monitoring** panel (**Out of Schedule** alert) are shown in orange.

The display contains the following data:

- **To WPT** – to select a WPT number for the display of associated schedule elements; the line to the right shows the name of the WPT if any was given by the operator;
- **Schedule** – to turn on the display of ETA and TTG calculated with regard to the prepared schedule;
- **Current SOG** – to turn on the display of ETA and TTG calculated with regard to the current speed (SOG);
- **ETA (UTC)** – estimated time of arrival in the selected WPT calculated from the remaining distance and set speed (UTC/Ship depending on the setting made in the Control panel);
- **TTG** – time to go to the specified WPT calculated with regard to the actual ETA;
- **DWOL** – distance to the wheel over line;
- **PTA** – planned time of arrival in the specified WPT taken from the schedule.

In the bottom part of the display there is a calculator for calculating speed to go (STG) to the specified waypoint or planned time of arrival (PTA) in the specified waypoint:

- **STG** – operator-entered planned speed of proceeding to the specified point for the PTA calculations in the window on the right;
- **PTA** – operator-entered planned time of arrival (PTA) to the specified point for the STG calculations in the window on the left.



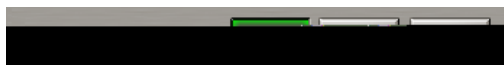
This page is intended for the control of chart automatic loading and scaling which are determined by the following parameters:

- **Chart autoload** – chart loading modes switching buttons:
 - **ON** – to turn on the chart autoload mode whereby the largest scale chart will be automatically loaded;
 - **OFF** – to turn off the autoload mode; as this is done, the current chart under the ship position or the chart loaded manually is fixed until the loading mode switch;
 - **FIX** – to turn on the chart fixing mode; as the ship symbol crosses the boundary of such chart, the chart loading mode is automatically switched to the ON position.
- **Chart autoscale** – chart scaling modes switching buttons:
 - **ON** – to load the chart on its original scale;
 - **OFF** – to load the chart on the screen current scale.
- **Scale ratio** – to set the loaded chart scale differing from its original scale by the number of fixed scale values (± 5);
- **Show only libraries for DNC charts** – to enable display of the DNC library name instead of DNC tiles names (if DNC charts used);
- **Chart priority** – to set the chart loading priority under the ship position:
 - **None** – to load charts most suitable for the set scale. In this case, if there are several charts on the same scale, priority is given to the TX-97 (TRX) format charts;
 - **ENC** – to load ENC format vector charts;
 - **DNC** – to load DNC format vector charts (not available for this version);
 - **HCRF** – to load ARCS and Seafarer format raster charts (not available for this version);
- **Navi-Planner** – to open the Navi-Planner 4000 application;
- **HCRF Review** – to open the window for viewing ARCS and Seafarer format charts (not available for this version).

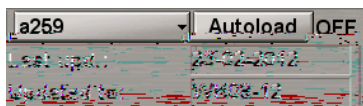
4.7.2.1. Chart Loading

To turn on chart autoloading:

1. Press the **ON** button in the **Chart autoload** row of buttons (if not depressed).



2. Press the **Autoload** button in the **Charts Area** window of the Control panel.



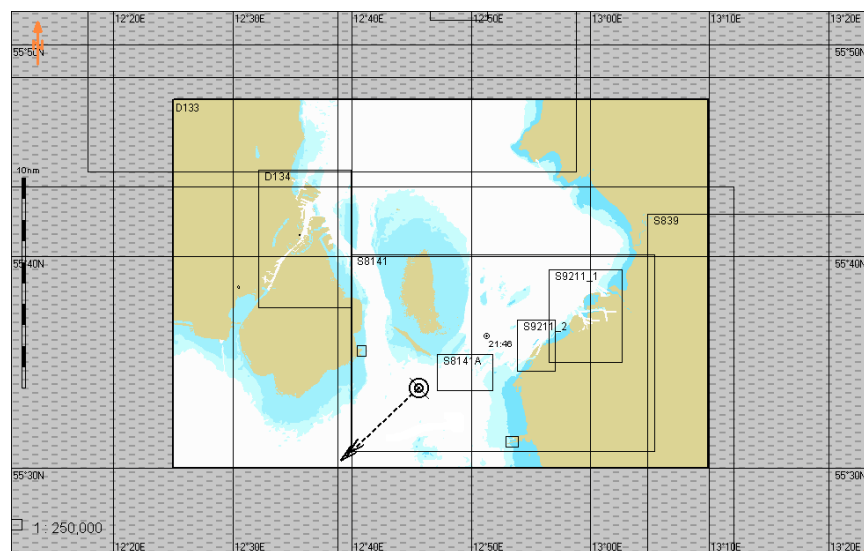
This window is designed for the adjustment of the chart display on the Chart panel:

- **Chart number** – to call a menu with a list of charts under the ship position (if the ship symbol is displayed on the chart) or the list of all the charts available in the ship folio. The button shows the current chart number;
- **Autoload** – to turn on the function for the automatic loading of charts. There is an indicator to the right of the button:
 - **ON** – the function is turned on (the **Autoload** button is greyed);
 - **OFF** – the function is turned off;
 - **FIX** – to fix the current chart.

The chart autoloading mode is set. For manual loading of the required chart, press the **OFF** button in the **Chart autoload** row of buttons (if not depressed).



The chart autoloading mode is OFF, and the screen displays the current chart only.



This page is intended for controlling display of individual chart information classes by using the following controls:

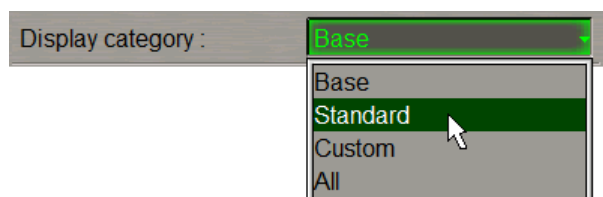
- **Display category** – to switch presentations of different display types:
 - **Base** – to turn on presentation of objects included in the Base Display;
 - **Standard** – to turn on presentation of objects included in the Standard Display;
 - **Custom** – to turn on presentation of objects included in the Additional Information Display at the user discretion;
 - **All** – to turn on presentation of all the objects included in the Additional Information Display.

According to the IEC 61174 standard, the following information has to be displayed during the route planning and monitoring:

- Display Base permanently shown on the ECDIS display, consisting of:
 - Coastline (high water);
 - Own ship's safety contour;
 - Isolated underwater dangers of depths less than the safety contour which lie within safe waters defined by the safety contour;
 - Isolated dangers above water which lie within the safe water defined by the safety contour such as fixed structures, overhead wires, etc;
 - Scale, range and north arrow;
 - Units of depth and height; and
 - Display mode.
- Standard Display, consisting of:
 - Display base;
 - Drying line;
 - Buoys, beacons, other aids to navigation and fixed structures;
 - Boundaries of fairways, channels, etc.;
 - Visual and radar conspicuous features;
 - Prohibited and restricted areas;
 - Chart scale boundaries;
 - Indication of cautionary notes;
 - Ship's routing systems and ferry routes;
 - Archipelagic sea lanes.
- All other information to be displayed individually on demand, for example:
 - Spot soundings;
 - Submarine cables and pipelines;
 - Details of all isolated dangers;
 - Details of aids to navigation;
 - Contents of cautionary notes;
 - ENC edition date;
 - Most recent chart update number;
 - Magnetic variation;

- Graticule;
- Place names.

Press the **Display category** button and select the **Standard** menu item from the drop-down list:



All the checkboxes for turning on the display of objects included in the display will be selected by default:

- **Buoys, beacons, aids to navigation** – to turn on the display of buoys, beacons, aids to navigation;
- **Boundaries and limits** – to turn on the display of boundaries and limits;
- **Prohibited and restricted areas** – to turn on the display of prohibited and restricted areas;
- **Ships' routing systems and ferry routes** – to turn on the display of ships' routing systems and ferry routes;
- **Archipelagic sea lanes** – to turn on the display of archipelagic sea lanes;
- **Cautionary notes** – to turn on the display of cautionary notes;
- **Drying line** – to turn on the display of drying lines;
- **Miscellaneous** - to turn on display of all the objects not covered by the **Standard** display;
- **Chart scale boundaries** – to turn on the display of chart scale boundaries;

Clear the appropriate checkboxes to turn off the display of unnecessary chart objects.

NOTE: If the Standard Display is selected, but some viewing group included in the Standard Display is turned off, the **STD DISP is customized** important indication is displayed in the **Operational** window of the Control panel (see also document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Display of Important Indications*).

There is another way to set the necessary layer.

The Standard Display can also be set by pressing the **STND DISP** button on the Control panel.



Settings on the **Layers** page and on the Control panel are equalised.

When the Standard Display is set, the display of the radar overlay, ARPA and AIS targets on the ECDIS task screen is automatically turned off.



To set displays from the keyboard:

- Press the <ALL LAYERS> key on the ES6 keyboard to turn on presentation of all the objects included in the Additional Information Display;
- Press the <STANDARD DISPLAY> key on the ES6 keyboard to set the Standard Display.

To turn on the display of the chart over the radar picture, press the  button in the Display Panel toolbar. In this case:

- Base Display mode is turned on;
- radar overlay is turned off;
- no targets (AIS, RIB, ARPA) are displayed;
- no tides or currents are displayed;
- no user chart objects are displayed;
- no Navtex objects are displayed.

To turn off Radar, AIS, ARPA overlays, press the  button in the Display Panel toolbar. In this case:

- Radar overlay is turned off;
- no targets (AIS, RIB, ARPA) are displayed

With the use of the ES6 keyboard, press the <SHOW CHART> key to activate the function.

Press the **Display category** button and select the **Custom** menu item from the drop-down list:



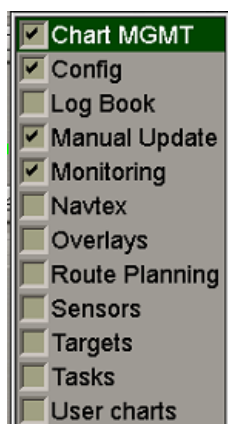
Select the checkboxes for turning on the display of necessary additional chart objects:

- All the objects included in the **Standard** display;
- **All Isolated dangers** – to turn on the display of isolated dangers with depths over them larger than the safety depth;
- **Submarine cables, pipelines** – to turn on the display of submerged cables and pipelines;
- **All depth contours** – to turn on the display of all the depth contours;
- **Tidal** - to turn on the display of tidal information;
- **Depth contours** – to turn on the display of depth contours with depths over them larger than the safety depth;
- **Seabed** – to turn on the display of ground characteristics;
- **Important text** – to turn on the display of important text;
- **Other text** – to turn on the display of other text (Names, Light description, etc.);
- **Unknown** - to turn on the display of objects which are not specified in S-52 standard;
- **Miscellaneous STD-** to turn on the display of all the objects not covered by the **Standard** display;
- **Miscellaneous** - to turn on the display of all the objects not covered by the **Custom** display;
- **Magnetic variation** – to turn on the magnetic variation;
- **Spot soundings to** – to turn on the display of depth values smaller than the setting made in the relevant window (if the setting is “0”, all the soundings are displayed).

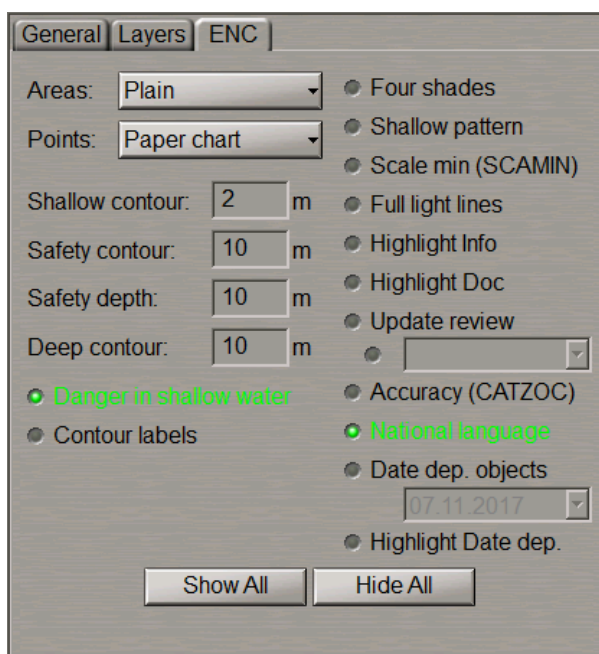
If the **All** menu item is selected from the **Display category** drop-down list, all the checkboxes for turning on the display of all available objects will be selected by default.

4.7.4. Handling ENC Format Charts

Open the **Chart MGMT** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the right-hand part of the **Charts** panel which will open up, to switch to the **ENC** page.



This page is intended for the adjustment and handling of the ENC format chart display.

From the **Areas** drop-down list, select the style of area type object boundary display (**Plain** and **Symbolised**).

From the **Points** drop-down list, select the style of the point type object display (**Paper chart** and **Simplified**).

Select the following checkboxes for turning on the display of necessary additional information classes:

- **Danger in shallow water** - to turn on the display of isolated danger objects which are located in the unsafe waters;
- **Contour labels** - to turn on the display of contour labels;
- **Four shades** – to turn on colour highlighting of the set depth areas;
- **Shallow pattern** – to turn on the display of hatching in the presentation of areas with depths smaller than the safety contour on the ENC format charts;
- **Scale min (SCAMIN)** – to turn on the display of an object on scales inadmissible for it (generalization method selection);
- **Full light lines** – to switch the lengths of lines delimiting the light visibility sectors;

5. Setting of Safety Parameters

5.1. Monitoring of Sailing along the Route and by the Schedule

ATTENTION! Setting of safety parameters in the sailing along the route and by the schedule is available only at the station with the MASTER status.

Open the **Monitoring** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.

- **End of route** – to enable alert generation as the last WPT of the monitored route is passed;
- **Out of XTL** – to enable alarm generation when the ship deviation from the route line is larger than the value set during the route planning;
- **Out of schedule** – to enable and set alerts generated when the ship is behind or ahead of the schedule;
- **WPT approach** – to enable alert generation a set period of time before the approach to the WPT;
- **Off leg course** – to enable and set parameters of the alert generated if the deviation between the current heading (HDG) and the route leg line direction exceeds the set value.

Select the **End of route** checkbox in the **Alerts** group to turn on the alert generation as the last point of the route is passed.

☒ End of route

The parameter of tracking the ship position relative to the current route leg is set by default. The XTL value is set at the time when the route is created on the **Route Planning** panel. The alarm is generated when the ship sails beyond the XTL limits.

☒ Out of XTL

Use the **WPT approach** line to enter the alert generation time value as the next WPT is approached. Select the activated **WPT Approach** checkbox to turn on the alert generated if the value of time of approach to the next WPT is less than the set one.

☒ WPT approach 5 min

Use the **Off leg course** line to enter the value of difference between the ship course and route leg direction. Select the activated **Off leg course** checkbox to turn on the alert generation as the difference between the courses exceeds the set value.

☒ Off leg course 5.0 °

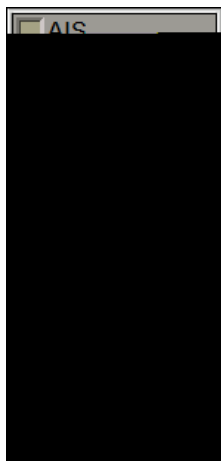
Use the **Out of schedule** line to enter the behind-the-schedule or ahead-of-the-schedule time relative to the loaded schedule. The alert is generated when the set value is exceeded. Select the activated **Out of schedule** checkbox.

☒ Out of schedule 10 min

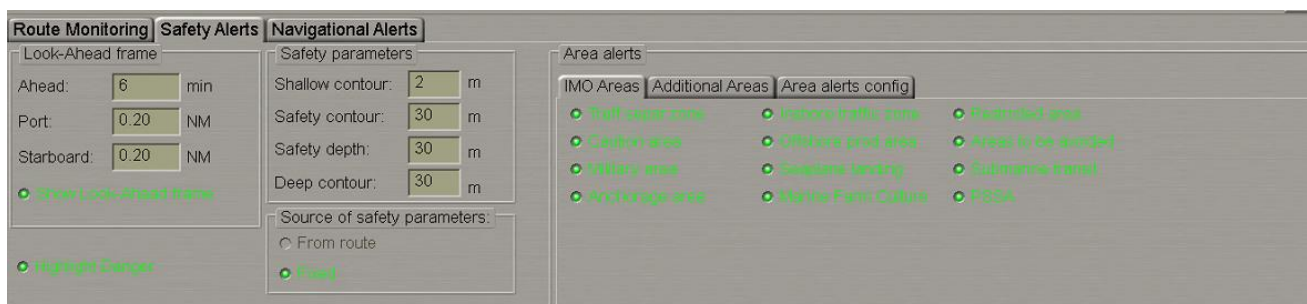
5.2. Monitoring of Safety Alerts

ATTENTION! Setting of safety parameters for the detection objects representing danger to navigation is available only at the station with the MASTER status.

Open the **Monitoring** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Monitoring** panel which will open up, to switch to the **Safety Alerts** page.



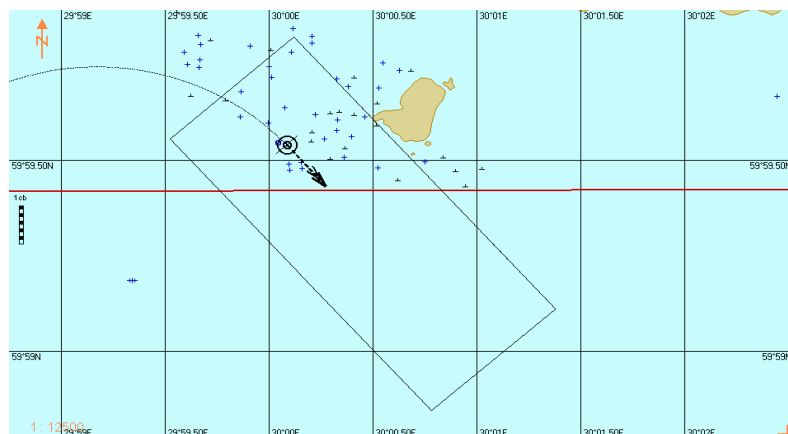
5.2.1. Setting of Safety Parameters for Detection of Dangerous Objects

The **Look-Ahead frame** group is intended for setting the size of the frame, which will be used for the chart data analysis and for the generation of the Antigrinding alerts, Area alerts and Navigational alerts:

- **Ahead** – window for the input of advance time for alert generation. The time value determines the length equal to the distance covered by the ship proceeding at the current SOG (from 30 s to 60 min);
- **Port** – to set the width of the corridor to the left of the ship (from 0.1 to 4.0 NM);
- **Starboard** – to set the width of the corridor to the right of the ship (from 0.1 to 4.0 NM);
- **Show Look-Ahead frame** – to turn on the display of a safety frame on the ECDIS task screen.

In the **Ahead**, **Port** and **Starboard** input boxes, set the necessary values which will determine the shape of the safety frame.

Select the **Show Look-Ahead frame** checkbox to show the safety frame. The safety frame is displayed with a black outlined rectangle:



The **Safety parameters** group is intended for setting and viewing safety parameters in the Navigation Mode:

- **Shallow contour** - in the ENC, the four shades mode defines the value of the shallow contour affecting colours of depth areas (dark blue for depth areas with values shallower than the shallow contour);
- **Safety contour** - mariner-selected contour to distinguish on the display between safe and unsafe water;
- **Safety depth** - mariner-defined depth to emphasize soundings on the display equal to or less than this value;
- **Deep contour** - in the ENC, the four shades mode defines the value of deep contour affecting colours of depth areas (the lightest S-52 colour for depth areas deeper than the deep contour).

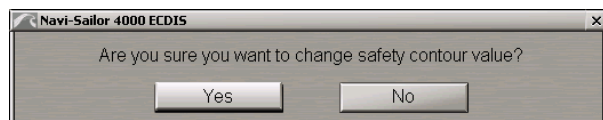
Use the **Shallow contour** line to enter the value of the shallow contour affecting colours of depth areas for ENC format charts:

Shallow contour: m

Use the **Safety contour** line to enter the safety contour value:

Safety contour: m

Press **Yes** to confirm the changes you have made:

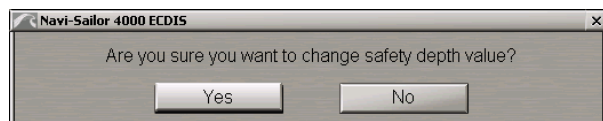


NOTE: The alert is generated exclusively by the safety contour value, even if the shallow contour value is smaller than the safety contour value.

Use the **Safety depth** line to enter the safety depth value:

Safety depth: m

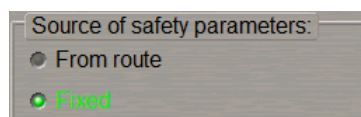
Press **Yes** to confirm the changes you have made:



Use the **Deep contour** line to enter the deep contour value affecting colours of depth areas for ENC format charts.

Deep contour: m

In the **Source of safety parameters** group, select the source which the safety parameters will be taken from for the route monitoring:



If the route is not loaded for monitoring, the **Fixed** radio button will be selected automatically, safety parameters set in the **Safety parameters** group are used for monitoring dangers.

As a route to be monitored is loaded, the last operator-selected option ('**From route**' or '**Fixed value**') is set automatically.

If the **Fixed** value is selected, the safety parameters set in the **Safety parameters** group are used (see also document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*).

If the **From route** value is selected, safety parameters are selected by using the procedure detailed in *Chapter 4, section Loading of Route and Schedule in the Navigation Mode, paragraph Selection of Safety Parameters when Loading Monitored Route*).

5.2.2. Setting of Safety Parameters in Sailing in the Vicinity of Dangerous Objects

NOTE: The **Antigrounding alerts** group contains checkboxes for enabling/disabling the following safety parameters excluded from the version 3.02.350.9047 and alerts by default in ON state

BEFORE	AFTER R10 3.02.350.9047
	

- **Safety contour** – to enable the alarm generation upon the approach to the depth line less than the **Safety contour** value;
- **Nav. hazard** – to enable the caution/warning generation upon the approach to the navigational hazards (for the list of objects see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*);
- **Highlight Danger** - to turn on the highlight the dangers on the Chart panel.

Select the **Safety contour** checkbox to turn on the alarm generation as the safety contour is crossed by the safety frame.

Select the activated **Nav. hazard** checkbox to enable the caution (warning) generation when a navigational hazard turns up within the safety frame (see also paragraph *Setting Priority for Navigational and Area Alerts*).

Select the **Highlight Danger** checkbox to turn on the highlighting of dangers on the Chart panel falling within the safety frame.

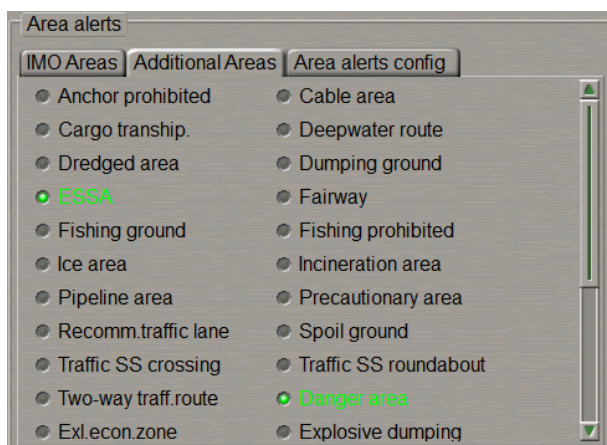
NOTE: If at least one checkbox in the **Antigrounding alerts** group is cleared, the Chart panel displays the **SF CNT** important indication (for details see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Indications, paragraph Important Indications*).

5.2.3. Monitoring of Crossing of Area Limits in IMO and Additional Groups

The **IMO Areas** tab containing checkboxes for the selection (assigning danger-to-navigation status) from the main areas tracked in accordance with IEC-61174 standard.



The **Additional Areas** tab containing checkboxes for the selection (assigning danger-to-navigation status) from the additional group of the areas that will be tracked.



Select the required areas checkboxes in the **IMO Areas** tab.

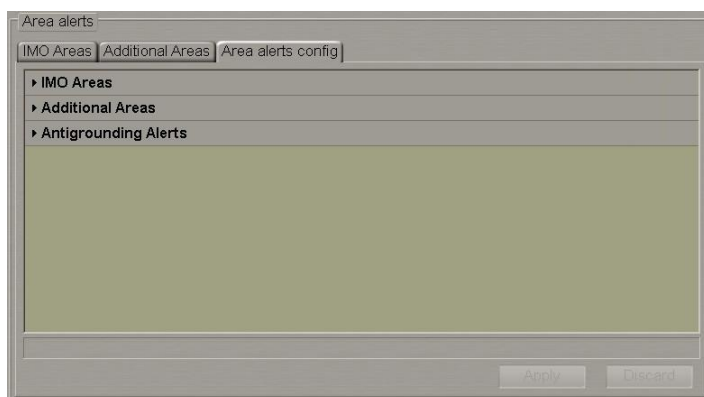
Switch to the **Additional Areas** tab. Select the required areas checkboxes in the **Additional Areas** tab.

NOTE: Priorities (caution/warning) for alerts caused by the safety frame crossing the selected areas are set on the **Area alerts config** tab (see also paragraph *Setting Priority for Navigational and Area Alerts*).

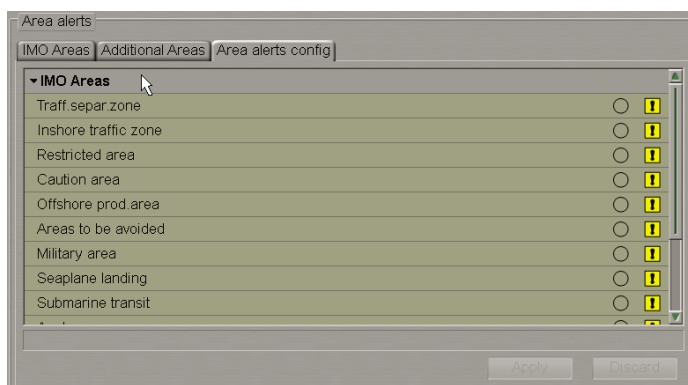
For details see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Main Alerts Generation Principles, paragraph Safety Alerts*.

5.2.4. Setting Priority for Navigational and Area Alerts

Switch to the **Safety alerts config** tab.



Click the button with the required list of alerts:

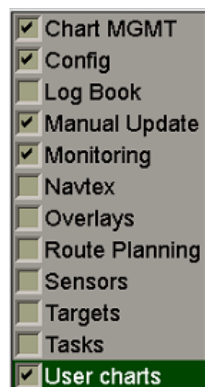


8. Handling User Charts

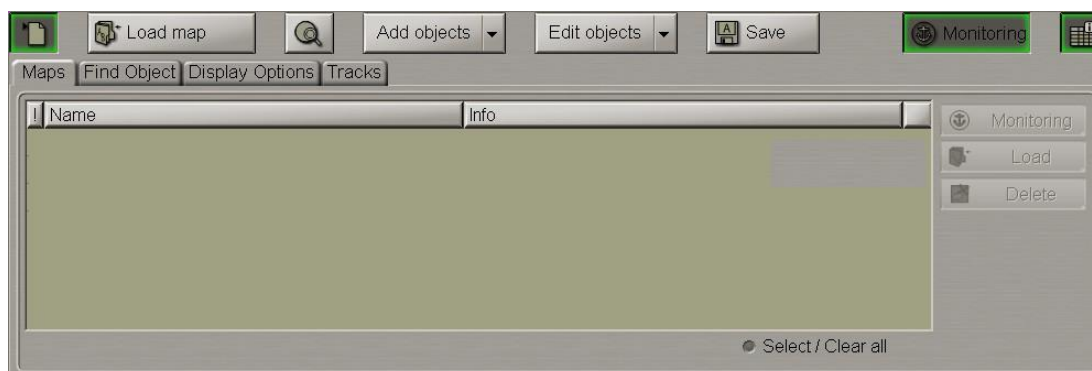
8.1. Creating and Saving User Charts

ATTENTION! Symbol type objects cannot be displayed on scales smaller than 1:600,000.

Open the **User Charts** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



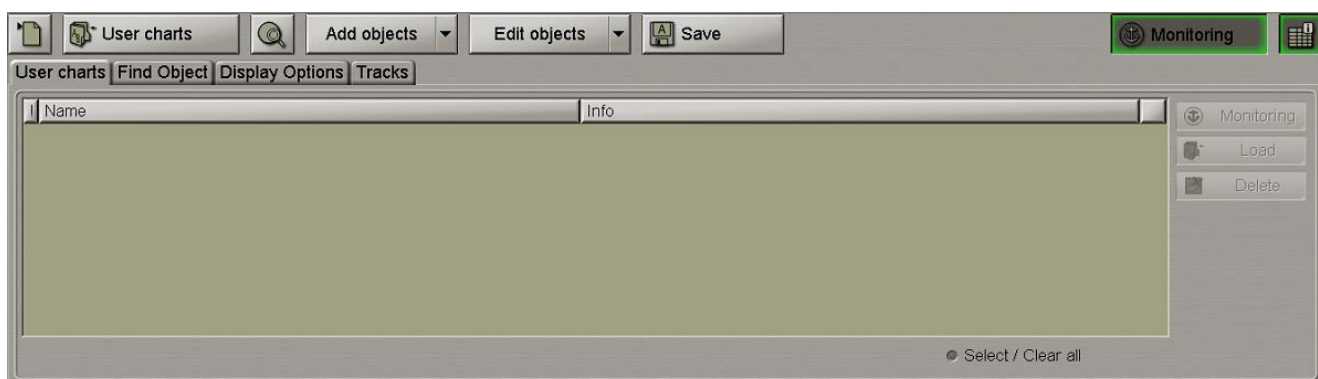
In the **User Charts** panel which will open up, press the  button in the top left part of the panel to turn on the user charts display on the ECDIS task screen.



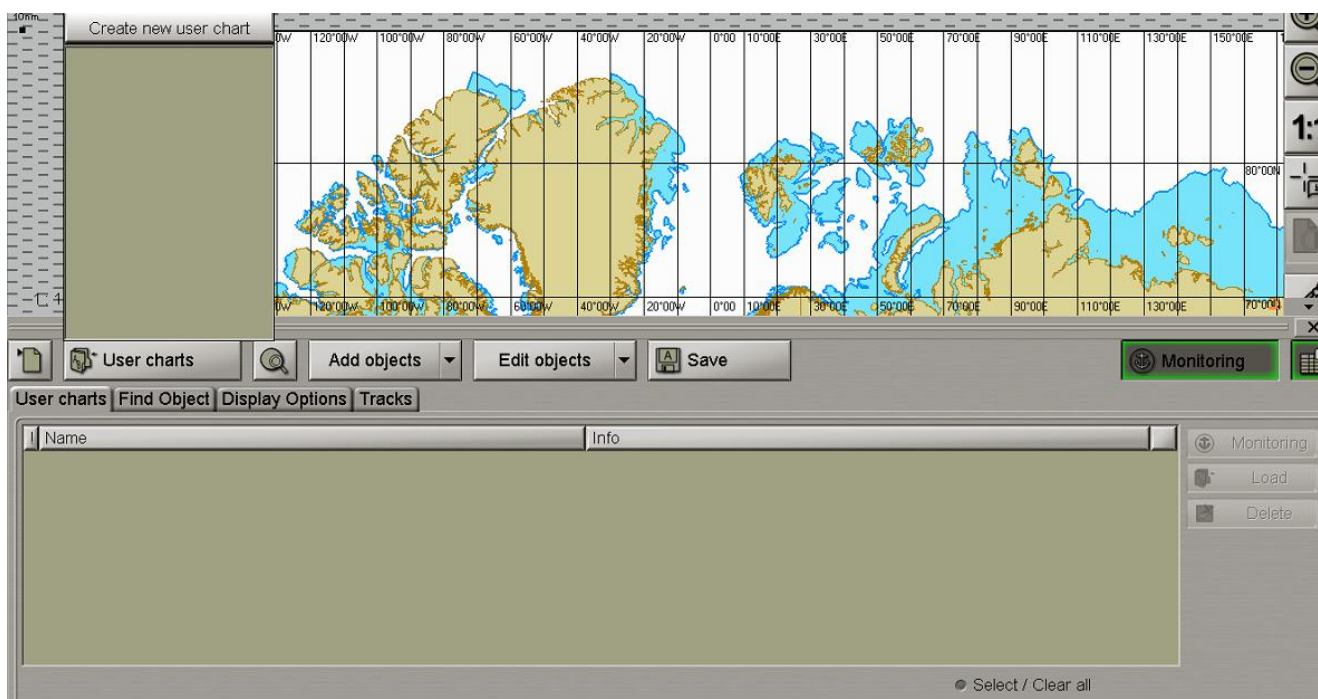
The user chart creating procedure consists of following stages:

- creation of user chart objects;
- selection of the object category and type;
- assigning of attributes to the object (except the "Attached files");
- plotting of an object;
- assigning of the file "Attached file" attribute
- saving the user chart.

Press the **User Charts** button.



The drop-down list displays all the loaded user charts.



In the drop-down list press the **Create new user chart** button.

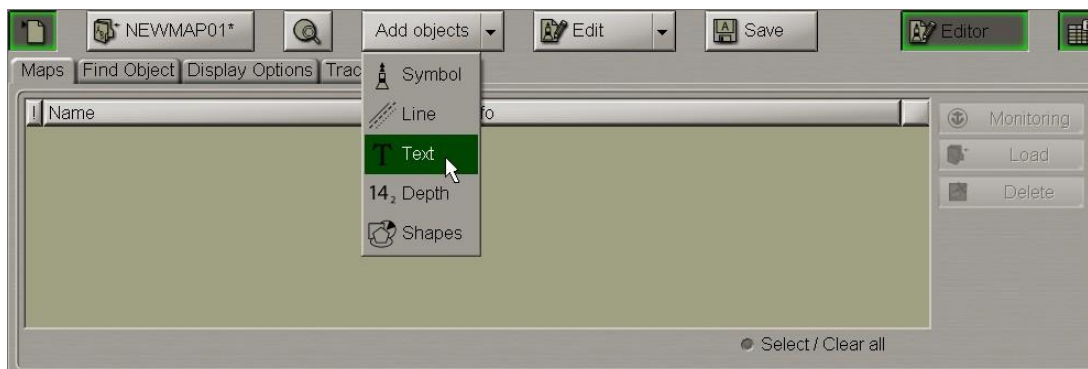


The user chart under the "NEWMAPXX*" name will be loaded.

NOTE: Next to the user chart name there is a "*" symbol which means that the user chart is edited but not saved. The unsaved are not synchronized with other WS's.

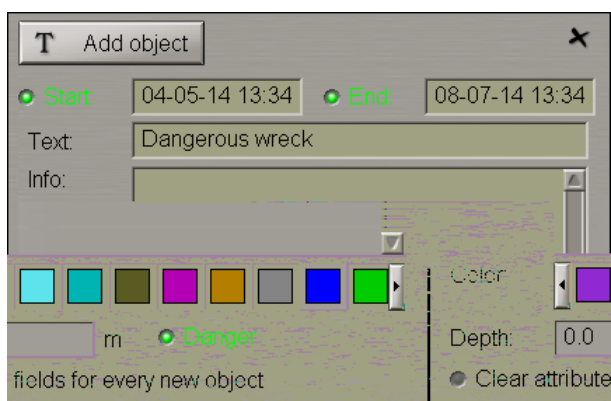
8.1.1.3. Plotting Text Object

Press the **Add objects** button.



From the drop-down list select the **Text** item.

In the **Text** input window enter the necessary text.



Set the user object colour ("Object colour" attribute), the user chart object displays time ("Temporary" attribute), required information ("Object information" attribute), the "Danger to navigation" attribute and the required depth value ("Object depth" attribute) (see item *Plotting Symbol Type Objects*).

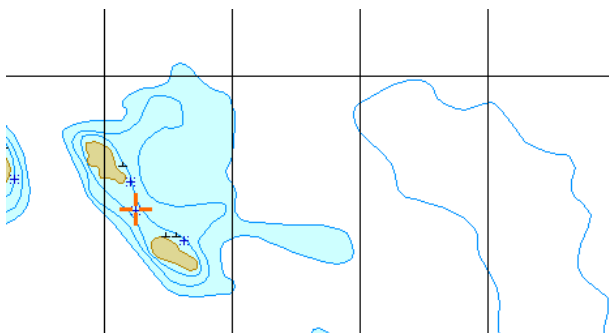
Files are attached ("Attached file" attribute) after the plotting of a user chart object.

To plot a user chart object on the chart, press the **Add object** button.

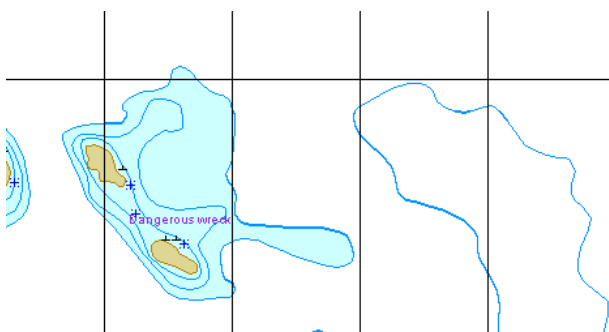
The graphics cursor  will appear on the Chart Panel, whereas the right bottom part of the screen will display the **Add Text** information window with cursor coordinates.



Position the cursor in the required coordinates.

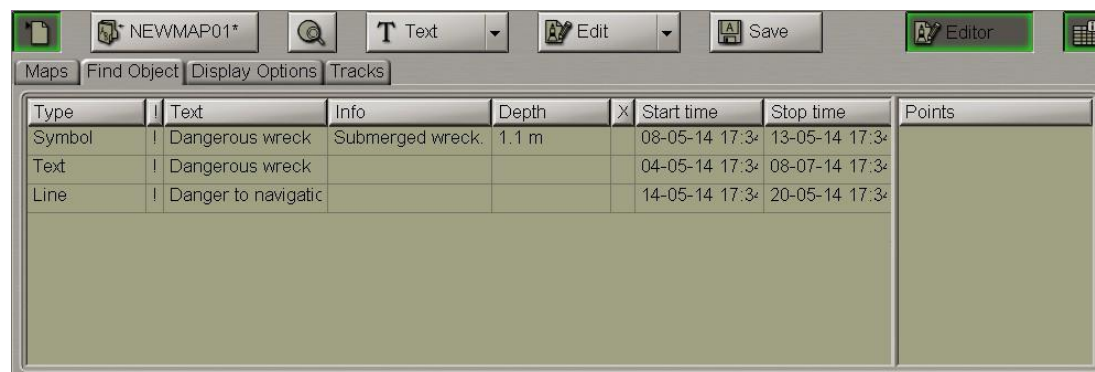


Press the left trackball button: the text type user chart object will be set in the specified coordinates.



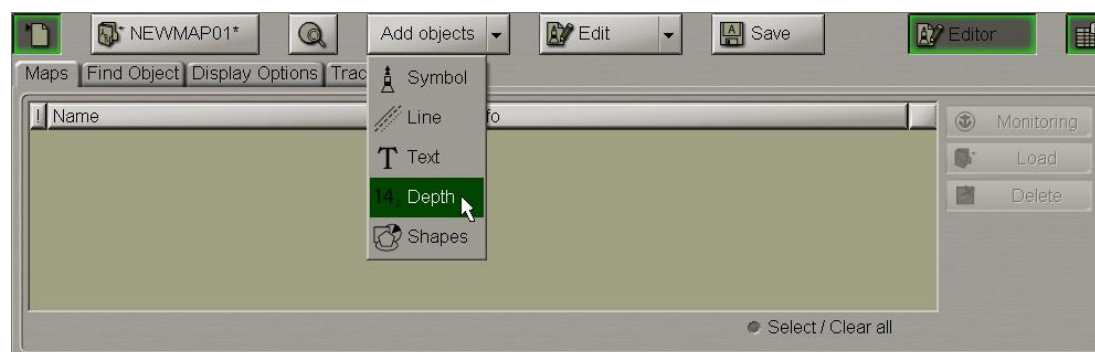
Press the right trackball button to exit from the editing mode. The cursor will assume the shape of the ECDIS task free cursor.

Switch to the **Find Object** page. Information on the plotted user chart object will be displayed in the table on this page.



8.1.1.4. Plotting Depth Object

Press the **Add objects** button.

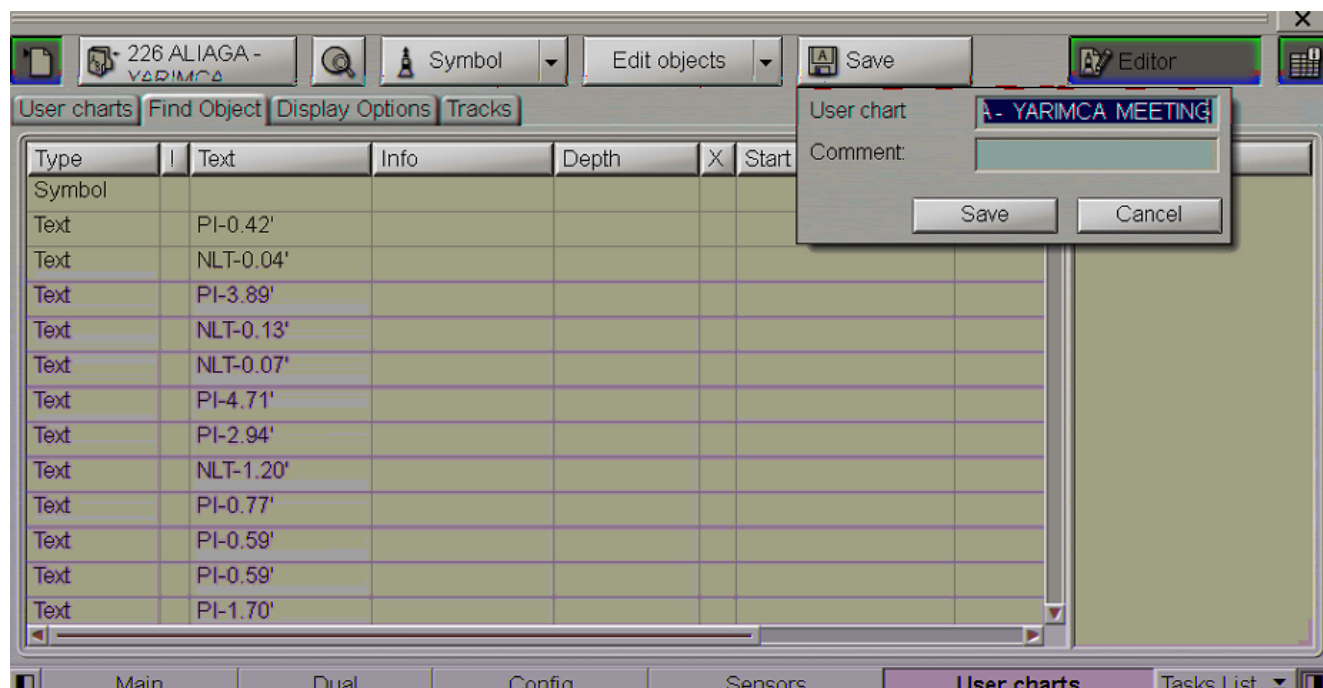


From the drop-down list select the **Depth** item.

8.1.2. Saving User Chart

Use the procedure detailed above to create and plot the required number of User chart objects.

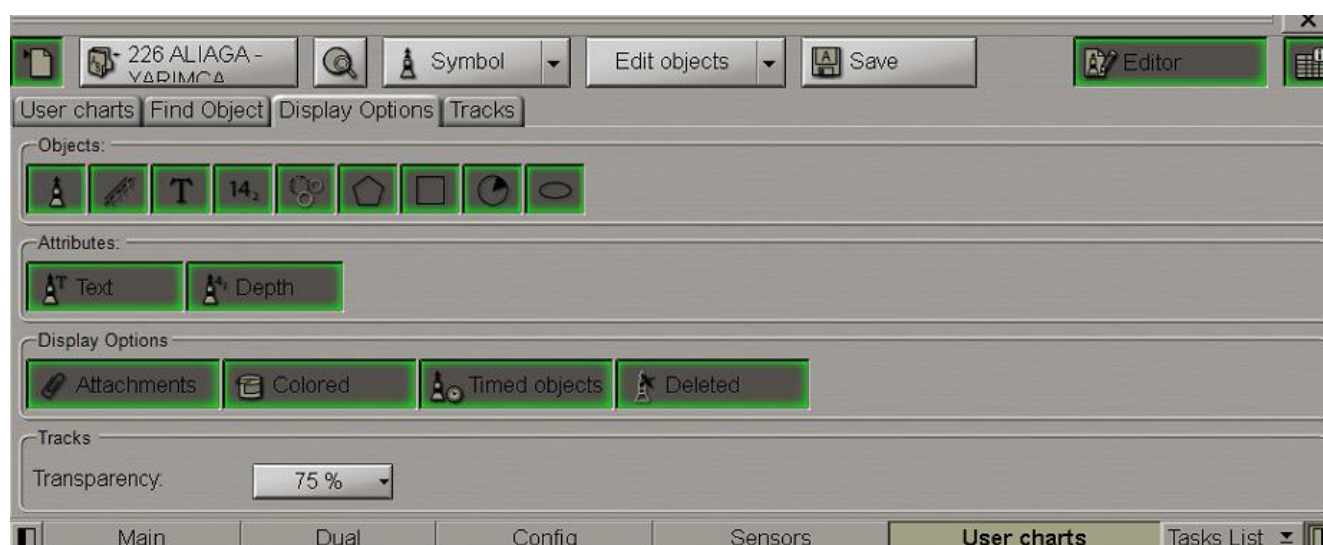
Press the **Save** button.



In the window which will open up, in the **Chart name** input line enter the User Chart name. In the **Comment** input line enter comments on the created user chart as required. Press the **Save** button in the window, the user chart will be saved.

8.1.3. Turning On/Off Display of User Chart Objects and Their Attributes

To turn on/off the display of user chart object categories on the ECDIS task screen, switch to the **Display options** page.

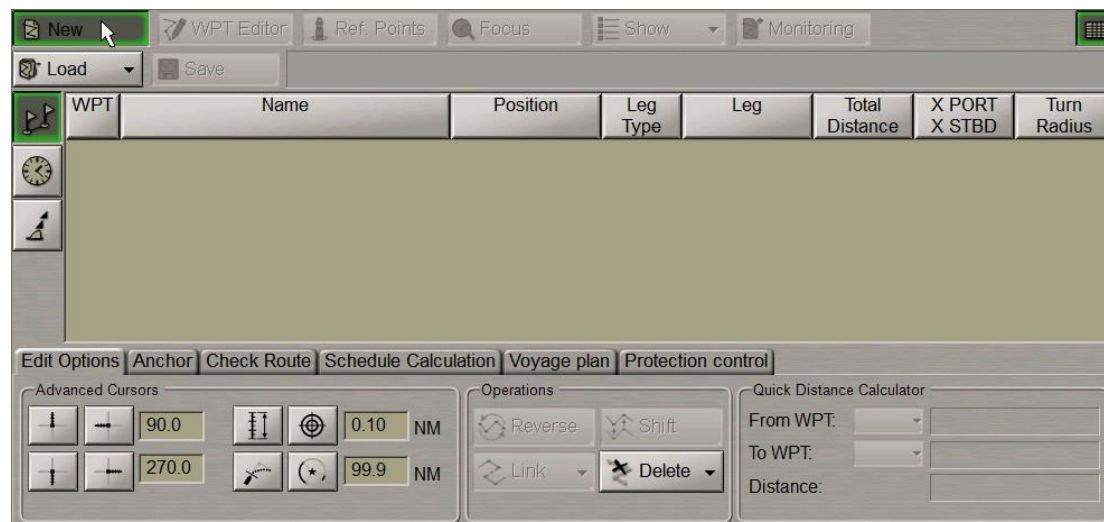


By default, all buttons are depressed.

To turn on/off the display of various user chart object categories on the ECDIS task screen, press the following buttons in the **Objects** group:

9.1.2.2. Plotting Route

Press the **New** button in the top left part of the panel to display a blank table for the generation of a new route.



The graphics cursor will appear on the Chart Panel, whereas the right bottom part of the screen will display the **Place First Point** information window.

Place first point

Latitude: 58°09.781' N

Longitude: 011°53.543' W

Anc. Brg: 000.0 °

Anc. Rng: 0.00 NM

Ship position:

true relative

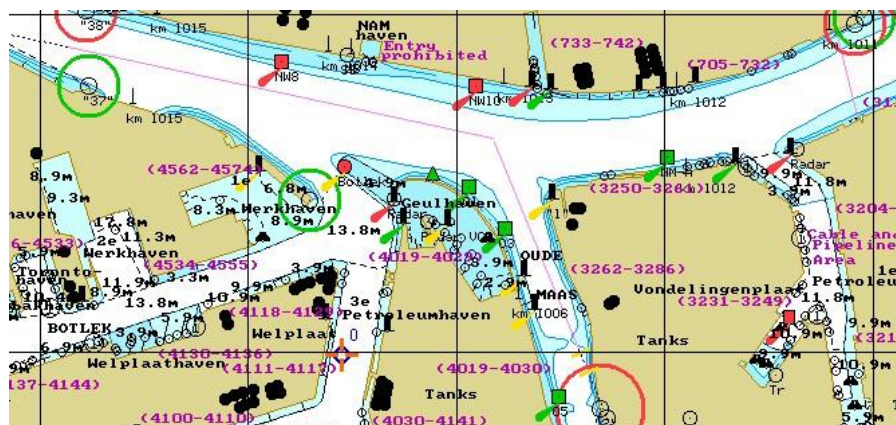
Bearing: 272.3 020.8 °

Opp. Bearing: 092.3 159.2 °

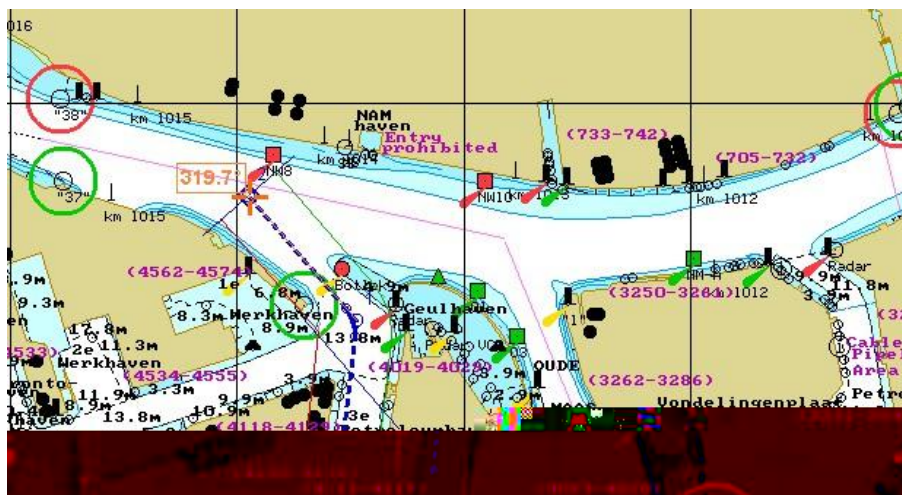
Range: 757.85 NM

Press Tab to edit manually

Set the cursor in the start point coordinates by moving the trackball, using the data in the **Place First Point** window, and press the left trackball button.

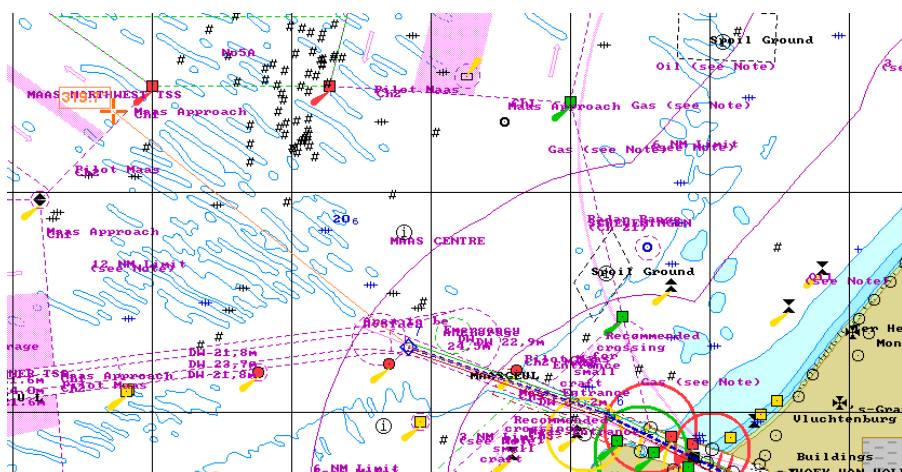


The symbol of the start point with its number ("0") will appear on the Chart Panel. Position the cursor in the coordinates of the next point. At this stage, a route leg line will be drawn between the cursor and the start point. Press the left trackball button: the Chart Panel will display the next point symbol with "1" for number and the plotted route leg. By default, XTD lines are also shown. To turn off their display, use the **Display** button (see below).

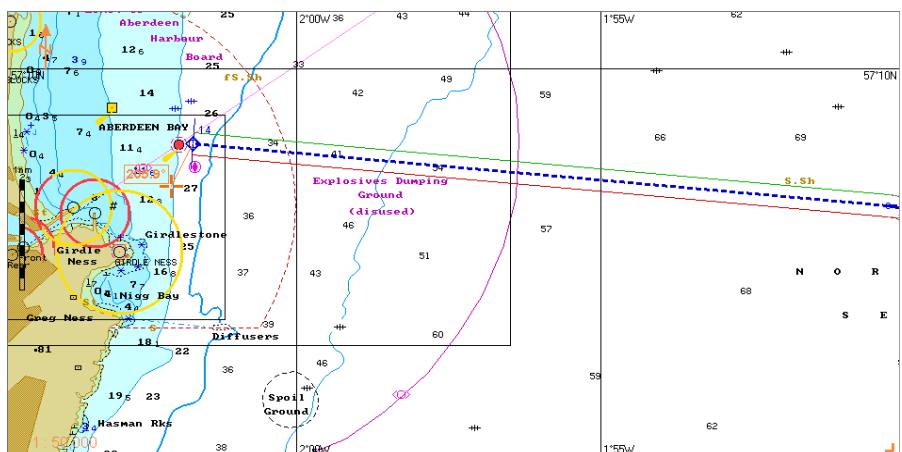


If it is necessary to set several waypoints, repeat the trackball moving and left button pressing operation as many times as there are WPT's required to be set.

If the cursor is positioned beyond the Chart Panel boundaries, it will be re-drawn automatically so that there is always a chart from the chart folio under the cursor.



After the setting of the last point of the planned route, double click the right trackball button: the cursor will exit from the route planning and editing mode and will assume the form of the free cursor.



The generation of the route is completed.

9.1.2.3.Entering Additional Data

The display of the table is turned on/off by using the **Show tables** button.



In the left part of the **Route Planning** panel, press the button to display the corresponding table.



New

WPT Editor

Ref. Points

Focus

Show

Monitoring

Load

Save

NONAME01*

	WPT	Name	Position	Leg Type	Leg	Total Distance	X PORT X STBD	Turn Radius
	0		51° 53.659 N 004° 18.420 E	XX	XXX.X XXX.X	XXX.X	XXX.X XXX.X	XXX.X
	1		51° 54.865 N 004° 14.838 E	RL	298.6° 2.27 NM	2.27 NM	0.10 NM 0.10 NM	0.50 NM
	2		51° 56.571 N 004° 11.176 E	RL	307.0° 2.84 NM	5.11 NM	0.10 NM 0.10 NM	0.50 NM
	3		51° 59.480 N 004° 02.548 E	RL	298.6° 6.09 NM	11.21 NM	0.10 NM 0.10 NM	0.50 NM

Edit Options

Anchor

Check Route

Schedule Calculation

Voyage plan

Protection control

Advanced Cursors

90.0

270.0

0.10

NM

99.9

NM

Operations

Reverse

Shift

Link

Delete

Quick Distance Calculator

From WPT:

0

To WPT:

15

Distance:

398.35 NM

The **Waypoints** table includes the following data (by columns):

- **WPT** – WPT number;
- **Name** – WPT name;
- **Position** – WPT latitude and longitude coordinate;
- **Leg Type** – route leg type (RL or GC) from the previous to the current WPT set by the operator; all route leg calculations are referenced to the WGS-84 reference datum;
- **Leg** – course and distance from the previous to the current WPT;
- **Total Distance** – summary distance between the initial (0) and current WPT by the route legs;
- **X PORT/X STBD** – safe zone to the left and to the right of the route leg;
- **Turn Radius** – operator-set radius of the arc for the display of an approximate ship track on the turn.

For input of the additional data, perform the procedures detailed below:

- position the cursor on the **Name** element cell; double click the left trackball button to activate the cell;
- enter the WPT name; press <ENTER>;
- position the cursor on the **Leg Type** element cell; double click the left trackball button to activate the cell;
- press the button in the right-hand part of the cell and select the sailing type from the list which will open up;
- press the left trackball button to confirm the changes you have made;
- position the cursor on the **XTL PORT/XTL STBD** element cell; double click the left trackball button to activate the cell;
- edit the XTD value to the left of the route leg; press the left trackball button;
- edit the XTD value to the right of the route leg; press the left trackball button;
- position the cursor on the **Turn Radius** element cell; double click the left trackball button to activate the cell;
- enter the radius for the arc of the suitable ship track on the turning circle; press the left trackball button.

Name

Name
St-Petersburg

Leg type
RL

Leg type
RL
RL
GC

Leg type
GC

X PORT
X STBD
0.10
0.10 NM

X PORT
X STBD
0.12 NM
0.12

X PORT	
X STBD	
0.12 NM	
0.12 NM	

Turn radius
0.10

Turn radius
1.00 NM

When filling in the table, you can use the context menu called by the right trackball button:

- **Clear cell** – to delete a value in the selected cell;
- **Fill cells down** – to fill in all the cells below with a value entered in the selected cell;
- **Fill cells up** – to fill in all the cells above with a value entered in the selected cell.

Total Time	Speed
	100%
	Copy waypoints
	Out waypoints
	Delete waypoints
	Insert waypoints
	Clear cell
	Fill cells down
	Fill cells up

In the left part of the **Route Planning** panel, press the **Extra Data** button to display the corresponding table.



WPT	Name	Remarks	Draught	UKC	Ref. Point	BRG/Dist	Safety contour
0			XXX.X	XXX.X			XXX.X
1			Undefined	Undefined			20 m
2			Undefined	Undefined			20 m
3			Undefined	Undefined			20 m

Below the table, there are sections for 'Advanced Cursors' (with values 90.0, 270.0, 0.10 NM, 99.9 NM), 'Operations' (Reverse, Shift, Link, Delete), and a 'Quick Distance Calculator' (From WPT: 0, To WPT: 88, Distance: 1374.50 NM).

For input of the additional route data, perform the procedures detailed below:

- position the cursor on the **Remarks** element cell; double click the left trackball button to activate the cell;
- enter the remark; press <Enter>;
- position the cursor on the **Safety contour** element cell; double click the left trackball button to activate the cell;
- edit the safety contour value for the current route leg; press the left trackball button (by default, and also for all the routes created in the earlier versions, the 30 m value is set);
- position the cursor on the **Safety depth** element cell; double click the left trackball button to activate the cell;
- edit the safety depth value for the current route leg; press the left trackball button (by default, and also for all the routes created in the earlier versions, the 30 m value is set).

Remarks

Remarks
Pilot disembarkation. Slow speed.

Safety contour
30

Safety contour
10 m

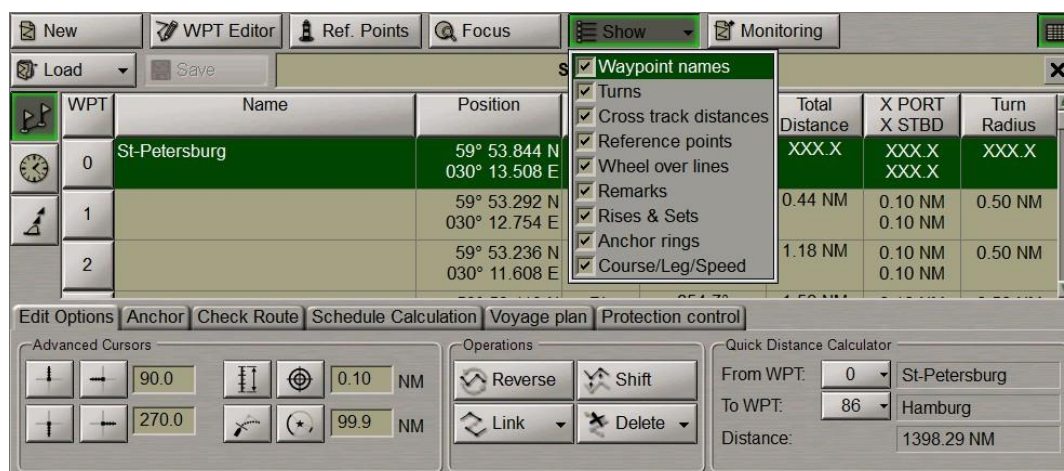
Safety depth
30

Safety depth
10 m

When filling in the table, you can use the context menu called by the right trackball button:

- Clear cell** – to delete a value in the selected cell;
- Fill cells down** – to fill in all the cells below with a value entered in the selected cell;
- Fill cells up** – to fill in all the cells above with a value entered in the selected cell.

Press the **Show** button in the top part of the panel.



The following checkboxes in the drop-down list for turning on the display of route options on the ECDIS task screen are selected by default:

- **Waypoints names** – to display on the electronic chart WPT names;
- **Turns** – to display on the electronic chart the turn radius arc for the active route;
- **Cross track distances** – to display on the electronic chart safety lines for the active route;
- **Reference points** – to display Reference points;
- **Wheel over lines** – to display on the electronic chart WOL marks;
- **Remarks** – to display remarks;
- **Rises & Sets** - to display sun and moon rise and set symbols;
- **Anchor rings** - to display anchor rings;
- **CRS/DIST/SPD** - to display course, distance and speed for each route leg.

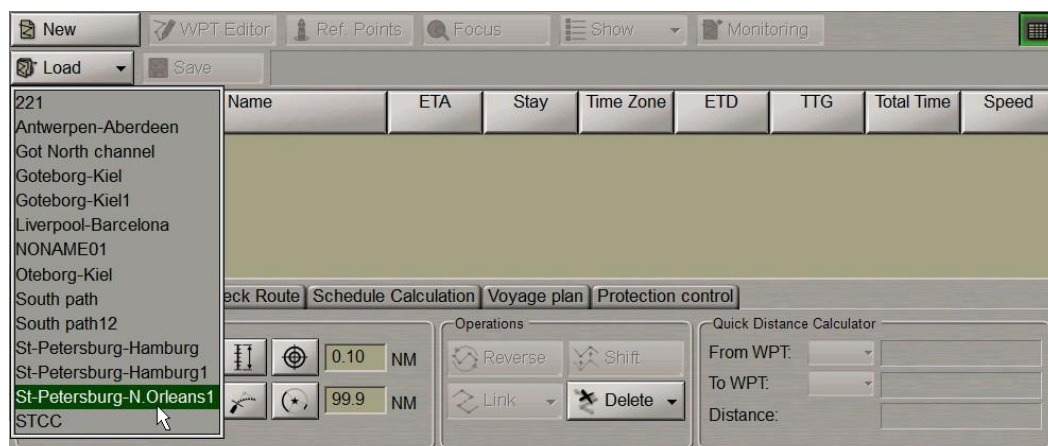
If necessary, clear the **Waypoints names** checkbox to turn off the display of waypoints' names on the ECDIS task screen.

To save the route, enter its name in the name input field and press the **Save** button.

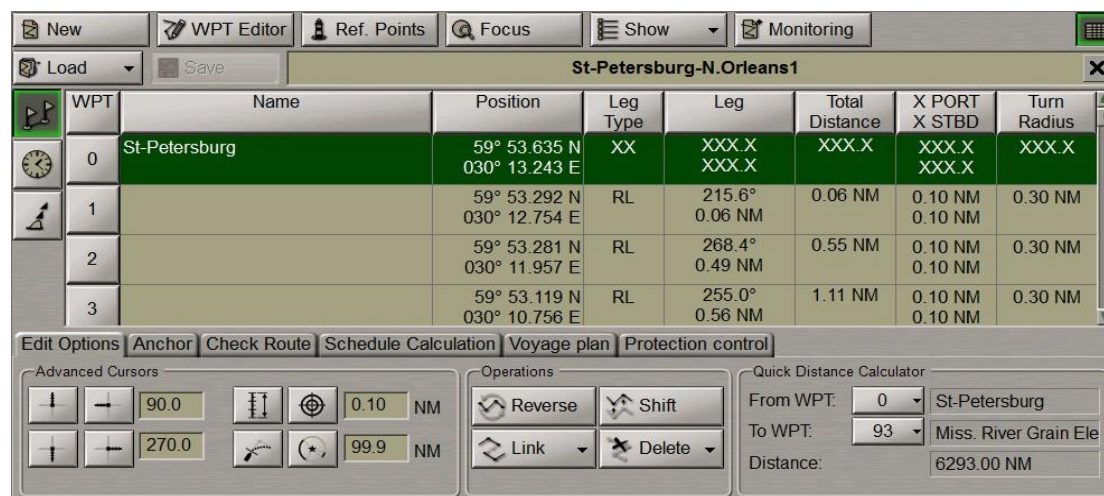


9.1.3.Route Loading and Unloading

In the top left part of the **Route Planning** panel, press the **Load** button.



In the list which will open up select the route required to be loaded in the editing mode and press the left trackball button.



Several routes can be loaded simultaneously in the editing mode. As a button with route name is pressed, the route becomes active and can be edited on the screen by using the graphic editor.



To quick load the necessary route in the monitoring mode, select it from the routes loaded in the editing mode and press the **Monitoring** button (see also *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). User Manual document, Chapter 4, Loading of Route and Schedule in the Navigation Mode* section and section *What is Different in Route Monitoring in the Route Planning* of this document):



To unload the route, press the  button to the right of the route name.

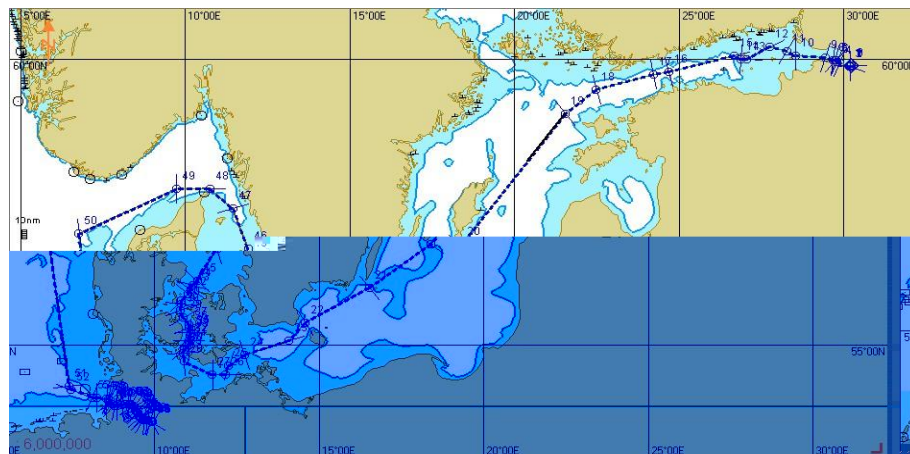
9.1.4. Fast Route Search

Load the necessary route (see item *Route Loading and Unloading*).

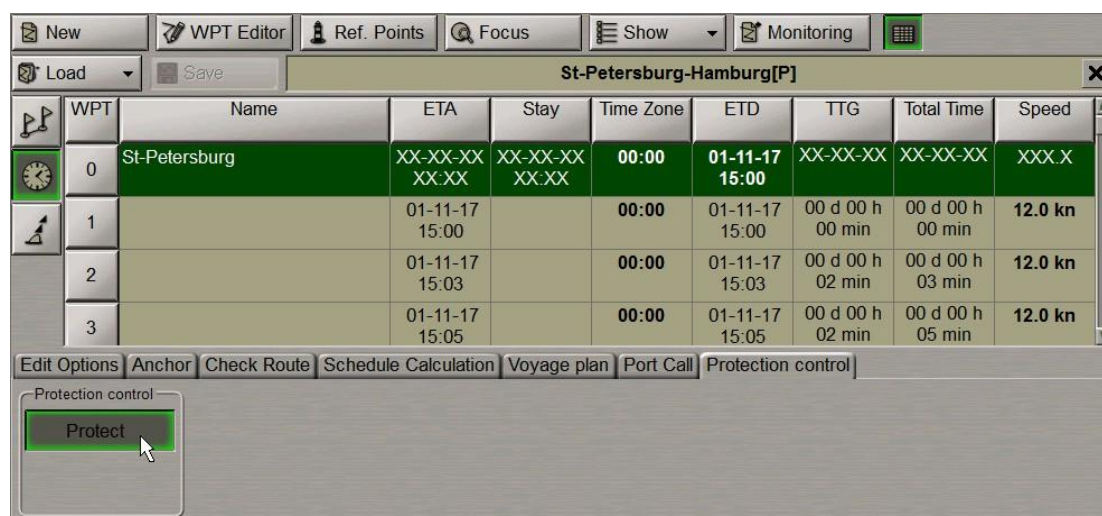
In the top part of the **Route Planning** panel, press the **Focus** button.



The Chart Panel will display the entire route on the suitable scale chart.

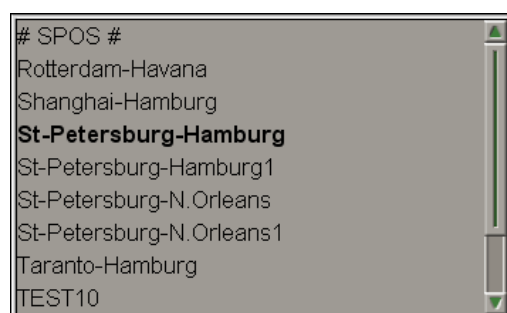


Enter the password for the appropriate access level. Press the **OK** button. The route will be protected from editing, and the letter “[P]” will be added to its name.



The protected route can neither be saved under the previous name, nor deleted. If saved under an altered name, the route loses its “Protected” status.

In the list of routes, the protected route is marked off with bold type:

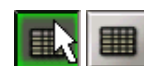


9.2. Work with Schedule

9.2.1. Drawing Schedule

Load the route for which the schedule will be drawn (see paragraph *Work with Route*, item *Route Loading and Unloading*).

The display of the table is turned on/off by using the **Show tables** button.



In the left part of the **Route Planning** panel, press the button to display the corresponding table.



In the left part of the **Route Planning** panel, press the button to display the corresponding table.



St-Petersburg-Hamburg

WPT	Name	ETA	Stay	Time Zone	ETD	TTG	Total Time	Speed
0	St-Petersburg	XX-XX-XX XX:XX	XX-XX-XX XX:XX	00:00		XX-XX-XX	XX-XX-XX	XXX.X
1				00:00				
2				00:00				
3				00:00				

Edit Options | Anchor | Check Route | Schedule Calculation | Voyage plan | Port Call | Protection control

Advanced Cursors: 90.0, 270.0, 0.10 NM, 99.9 NM

Operations: Reverse, Shift, Link, Delete

Quick Distance Calculator: From WPT: 0 St-Petersburg, To WPT: 86 Hamburg, Distance: 1398.25 NM

To draw a new schedule, switch to the **Schedule calculation** page.

Edit Options | Anchor | Check Route | Schedule Calculation | Voyage plan | Protection control

Schedule calculation

Create Schedule, Clear Schedule, Delete Schedule

Press the **Create Schedule** button in the **Schedule** group.

Schedule

Create Schedule, Clear Schedule, Delete Schedule

Enter schedule elements for WPT's in the table. To do this, perform the procedures detailed below:

- position the cursor on the **ETA** element cell (not set for the route start point). Double click the left trackball button to activate the cell;

ETA

23-09-09
19:18

- enter the ETA for this WPT and press the right trackball button;

ETA

29-09-09
19:00

Apply
Cancel

- press the **Apply** button to confirm the changes you have made;

ETA

29-09-09
19:00

- position the cursor on the **Stay** element cell; double click the left trackball button to activate the cell;

Stay

00 d 00
h 00 m

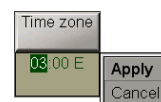
- enter the time of stay at this WPT and press the right trackball button;

Stay

00 d 05
h 00 m

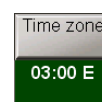
Apply
Cancel

- press the **Apply** button to confirm the changes you have made;
- position the cursor on the Time zone element cell; double click the left trackball button to activate the cell;

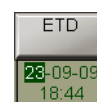


- enter the time zone at this WPT and press the right trackball button;

- press **Apply** button to confirm the changes you have made;

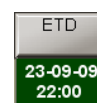


- position the cursor on the **ETD** element cell (not set for the route end point); double click the left trackball button to activate the cell;

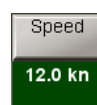


- enter the **ETD** for the WPT and press the right trackball button;

- press the **Apply** button to confirm the changes you have made;



- position the cursor on the **Speed** element cell; double click the left trackball button to activate the cell;

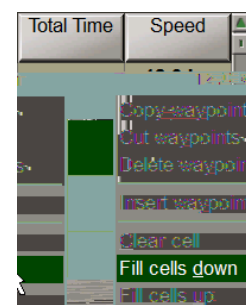


- enter the speed value on the route leg; press the left trackball button.

NOTE: There is no editing of **TTG** and **Total Time** cells.

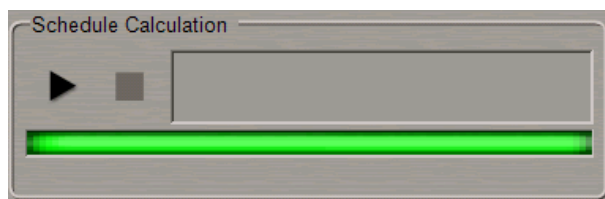
When filling in the table you can use the context menu called by the right trackball button:

- **Clear cell** – to delete a value in the selected cell;
- **Fill cells down** – to fill in all the cells below with a value entered in the selected cell;
- **Fill cells up** – to fill in all the cells above with a value entered in the selected cell.



ATTENTION! For the route start point, the ETD value should always be specified. For the end point, it is necessary to specify ETA or speed of proceeding along the route legs.

Press the  button in the **Schedule Calculation** group: the schedule will be calculated.

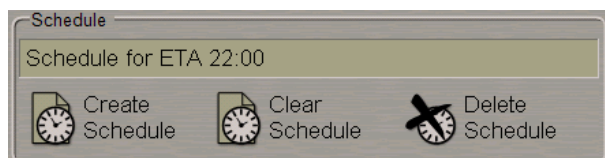


The schedule calculation process is reflected in the display window. In the table, empty cells will be filled in with the calculated schedule elements.

WPT	Name	ETA	Stay	Time Zone	ETD	TTG	Total Time	Speed
0	St-Petersburg	XX-XX-XX XX:XX	XX-XX-XX XX:XX	03:00 E	07-11-17 18:00	XX-XX-XX	XX-XX-XX	XXX.X
1		07-11-17 15:00		00:00	07-11-17 15:00	00 d 00 h 00 min	00 d 00 h 00 min	12.0 kn
2		07-11-17 15:03		00:00	07-11-17 15:03	00 d 00 h 02 min	00 d 00 h 03 min	12.0 kn
3		07-11-17 15:05		00:00	07-11-17 15:05	00 d 00 h 02 min	00 d 00 h 05 min	12.0 kn

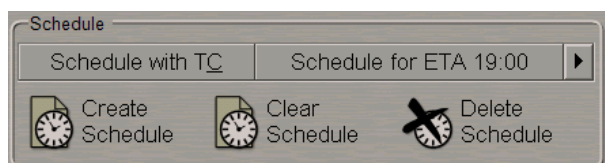
Below the table, the **Schedule Calculation** window shows "Calculation completed" and a progress bar. The **Schedule** window shows "Schedule 1" and buttons for "Create Schedule", "Clear Schedule", and "Delete Schedule".

A schedule has been created. To save the schedule, enter its name in the name input field of the **Schedule** group.



Press <ENTER>.

You can simultaneously create several schedules with different data for one and the same route. As a button with schedule name is pressed, the schedule becomes active and can be edited in the schedule Table. The active schedule can be selected in the line of the **Schedule** group:



10. Handling Radar Information and Target Designation Units

10.1. Radar Overlay

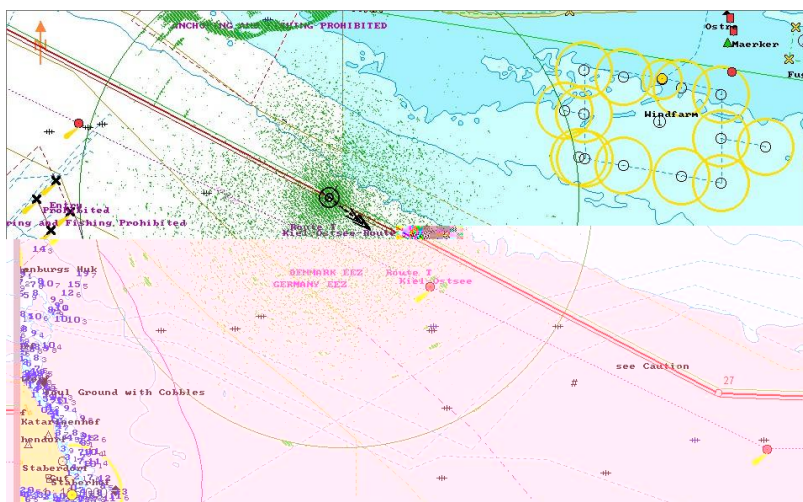
To select a scanner, press the scanner selection button arrow in the **Sensors** window of the Control panel.



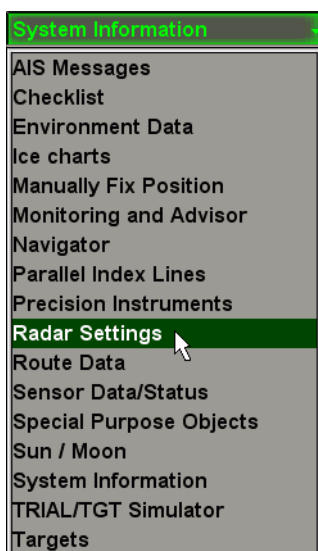
Select the necessary scanner (the button name corresponds to the selected scanner). To turn on the radar picture, press the **Overlay** button.



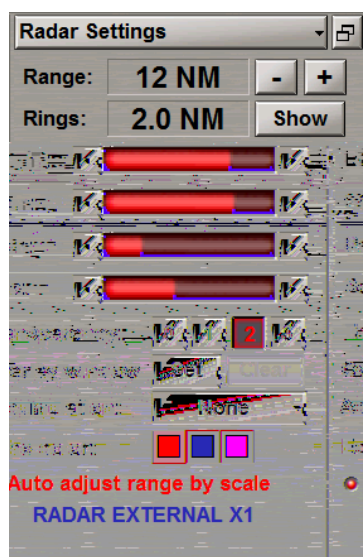
A radar picture will appear on the Chart panel.



Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Radar Settings** line and press the left trackball button.



In the **Radar Settings** display which will open up, set the range scale. To do this, use the buttons to the right of the **Range** input box.



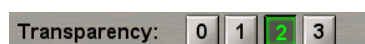
Turn on the display of range rings as required. To do this, press the **Show** button to the right of the **Rings** box.



Set the required brightness, gain, suppression of rain and sea clutter by using the buttons on the sides of level indicators.



To enter the required transparency value, use the buttons (from 0 – opaque to 3 – almost transparent) in the **Transparency** group.



To set the required rectangular area of the radar picture display, press the **Set** button in the **Overlay window** group.



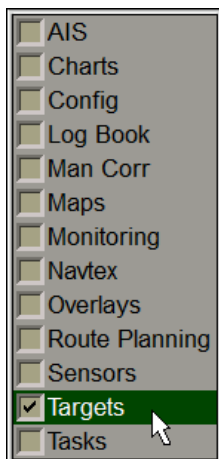
A graphics cursor will appear in the chart screen area. Position the cursor in the initial point of the area and press the left trackball button. Set the required radar picture area by moving the cursor, keeping the left trackball button depressed.

10.2. ARPA

10.2.1. Target Display

There are three ways to turn on/off the display of ARPA targets on the ECDIS task screen:

1. Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



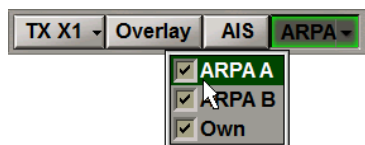
Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

Target Table		AIS color tracks												
Show Target		ARPA A Own AIS Tracks ARPA B Cursors ERBL												
		AIS Target identification By ID Max. range 48 nm												
		Alarm CPA 1.00 NM TCPA 1.0 min												
Name	40	41	42	43	44	37	47	48	49	50	51	52	53	54
Alias														
CPA (NM)	2.88	5.87	7.32	6.81	7.74	5.81	4.09	4.53	5.86	8.23	1.25	6.76	8.07	5.62
TCPA (min)	-9.7	-8.4	-6.9	-14.5	-9.3	-21.2	-21.2	-21.2	10.4	12.8	21.8	-16.4	-16.6	-19.2
SOG (kn)	45.3	43.5	29.4	26.8	33.1	11.3	15.0	13.1	22.4	12.3	16.8	19.5	15.8	12.8
COG (°)	14.5°	295.9°	65.7°	287.1°	281.1°	30.0°	5.9°	36.4°	84.2°	170.7°	81.6°	348.7°	358.2°	29.6°
Range (NM)	8.94	8.33	8.43	9.10	9.05	8.79	8.67	8.57	7.63	8.46	8.44	9.59	10.11	8.56
Bearing (°)	359.3°	350.7°	0.3°	351.7°	352.5°	352.7°	45.2°	5.8°	203.4°	236.6°	257.5°	317.5°	318.4°	352.4°
Bow X (NM)	7.7	-5.9	-20.9	-6.9	-8.1	55.4	-10.5	112.1	10.6	8.2	-2.6	11.1	16.8	49.3
TBow X (min)	-17.9	-9.0	25.4	-16.5	-13.8	-197.9	5.6	-346.8	-8.4	15.6	27.7	-37.6	-56.9	-164.8

In the **Show Target** group, press the following buttons:

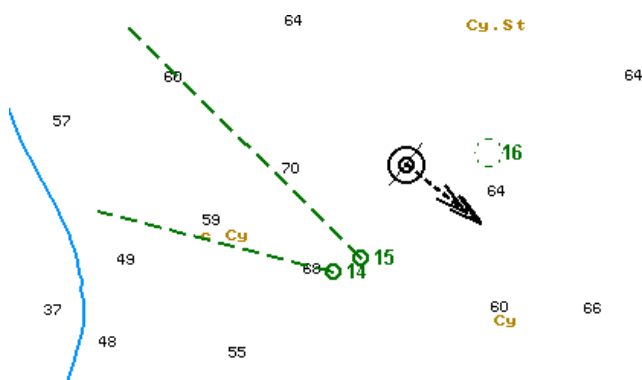
- **ARPA A(B)** – to turn on the display of ARPA A(B) targets. Connection is made in the System Configuration utility (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Installation Guide, Chapter 2, section MFD Configuration, paragraph Sensors Settings*);
- **Own** – to turn on the display of ARPA targets processes by the ARPA built into the program.

2. In the **Sensors** window of the Control panel select the necessary source of ARPA targets and press the **ARPA** button. From the drop-down list select the necessary ARPA source.



3. Press the <TARGET> key on the ES6 keyboard (the display of all the ARPA targets from all the available sources is turned on).

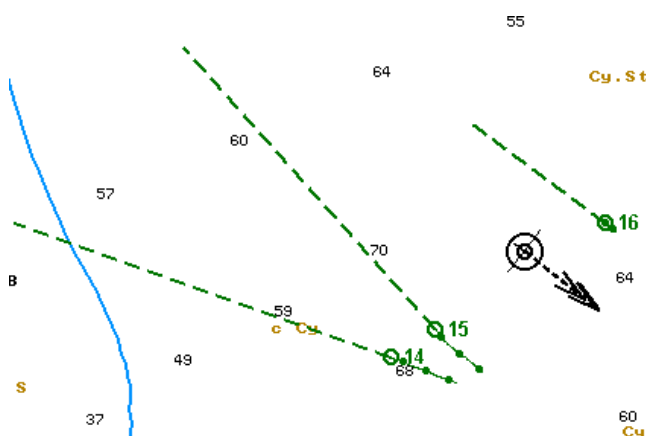
If there are some ARPA tracked targets, they will appear on the Chart panel, whereas their motion parameters will be shown in the target table.



Turn on, as required, the display of target tracks. To do this, press the **Tracks** button in the **Show Target** group.



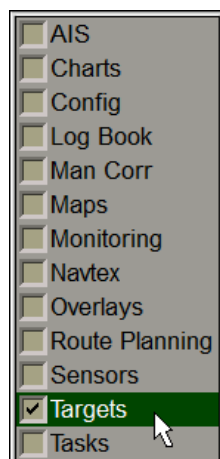
Target tracks will be displayed on the Chart panel.



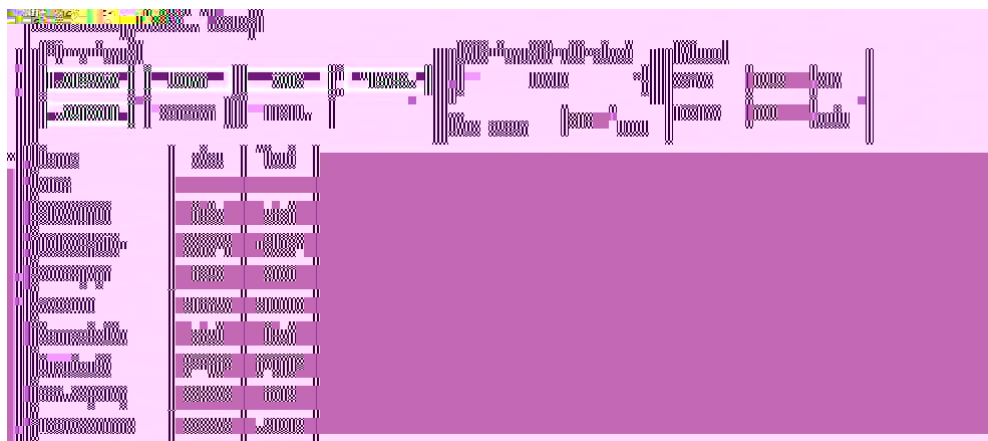
10.2.2. Turning On/Off External ARPA Tools

ATTENTION! The tools are not available unless the external ARPA-A is connected and provides relevant sentences.

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.

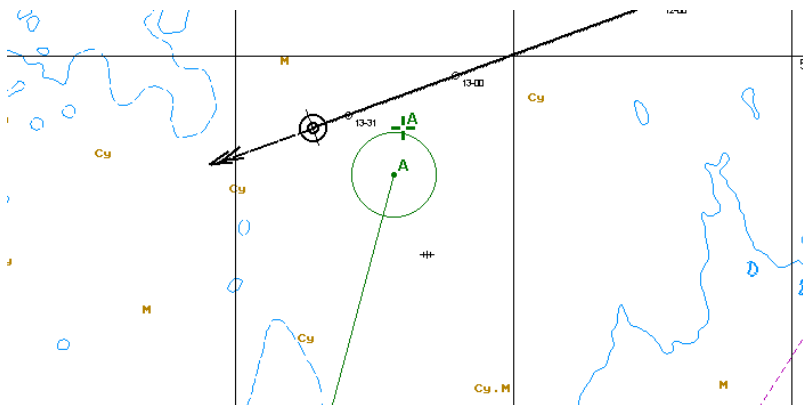


Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:



In the **Show Target** group of the **Targets** panel which will open up, press the **ARPA A** button.

To turn on the display of tools on the Chart panel, press the **Cursors** and/or **ERBL** buttons in the **Show Target** group.

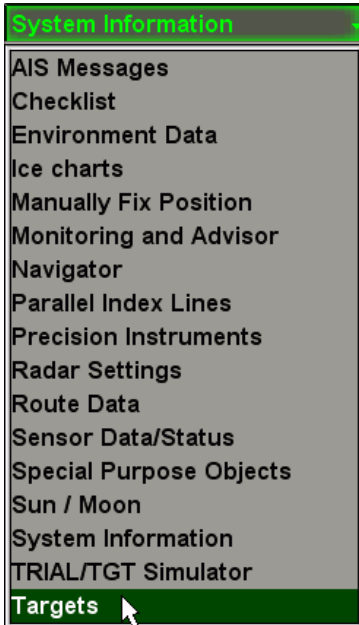


10.2.3. ARPA Settings

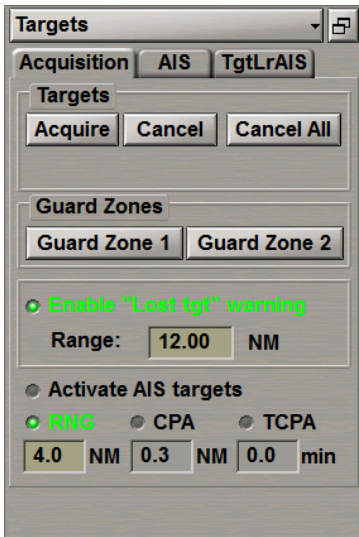
To turn on the display of targets, press the **Overlay** and the **ARPA** buttons in the **Sensors** window of the Control panel.



Press the button with the name of the set display in the **Display Panel** window of the Control panel.



In the list which will open up, select the **Targets** line and press the left trackball button.

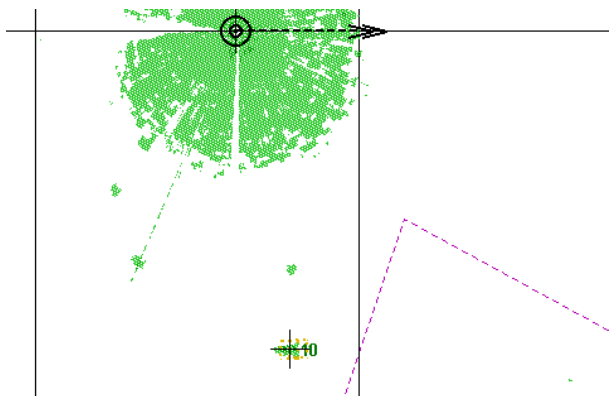


10.2.3.1. To Acquire Targets for Tracking

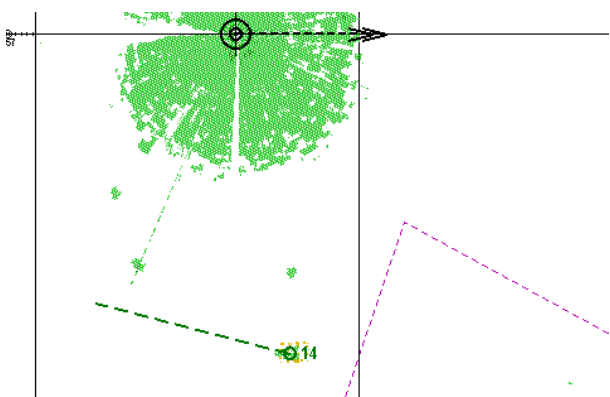
To switch the cursor to the target acquisition mode, press the **Acquire** button in the **Targets** group.



Position the cursor on the target and press the left trackball button.



In a minute, the acquired target will be displayed as a green coloured circle with pre-calculated motion vector.



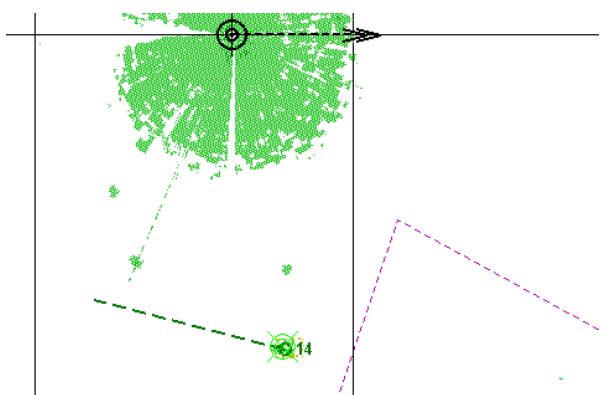
The target has been acquired for tracking. If you press the right trackball button, the cursor exits from the target acquisition mode.

10.2.3.2. To Cancel Tracked Targets

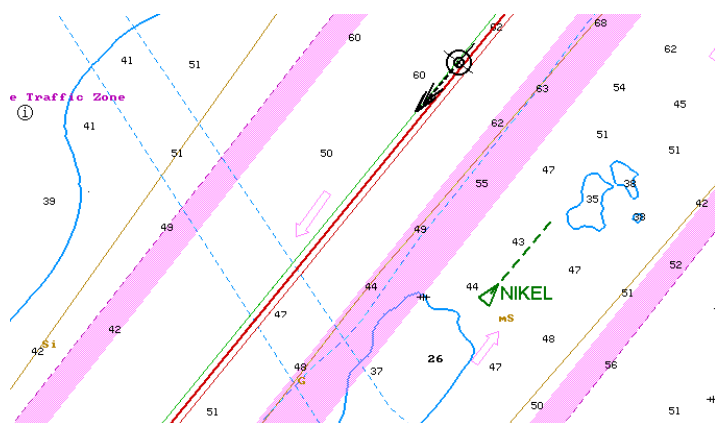
To switch the cursor to the target tracking cancelling mode, press the **Cancel** button in the **Targets** group.



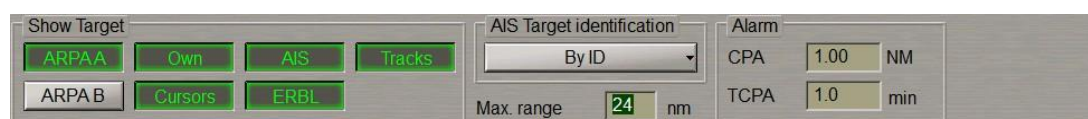
Position the cursor on the tracked target and press the left trackball button.



AIS targets will appear in the Chart panel, and their motion parameters will be displayed in the Targets Table.



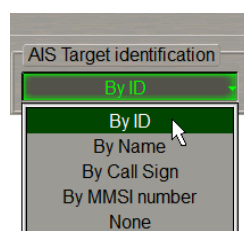
48 nm value is set in the **Max. Range** input box by default, that is the radius of an area within which the AIS targets are processed and displayed.



Set the necessary radius (up to 100 NM) of the AIS targets processing area.

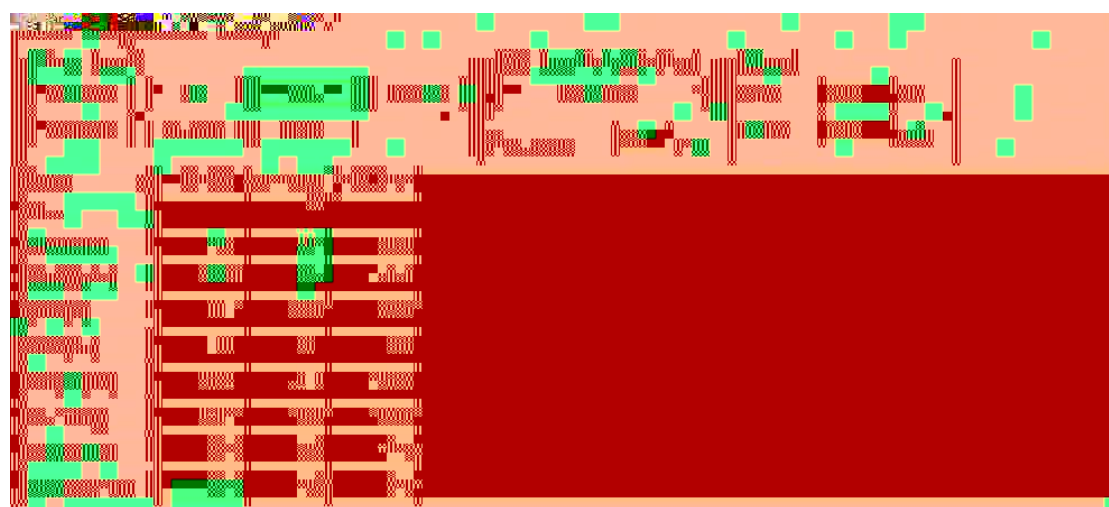
10.3.1.2. Display of AIS Target Identifiers

Press the **AIS Target identification** button to select the AIS targets identifier.



In the list which will open up, select the necessary identifier or **None** to declutter the Chart panel. Press the left trackball button. Targets will be displayed on the Chart panel with the selected identifier, and its name will be shown on the button.

Position the cursor on the target identifier in the Targets Table.

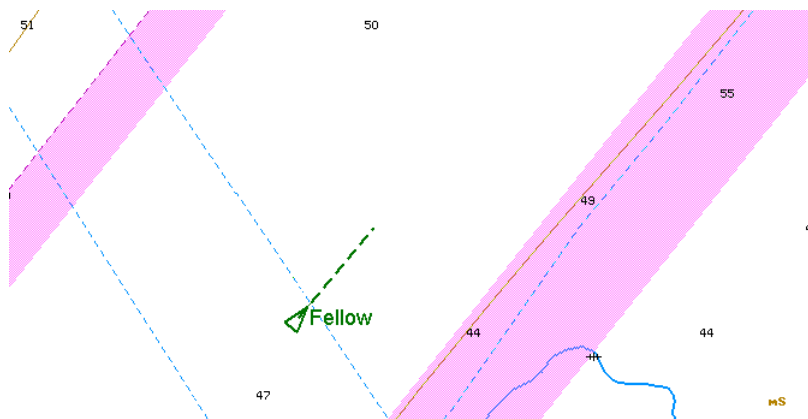


The hint will display the name of the AIS object if it is contained in the messages transmitted by the AIS object.

Enter as required the “Alias” identifier (up to 8 characters) for an AIS object in the **Alias** row of the Targets Table.

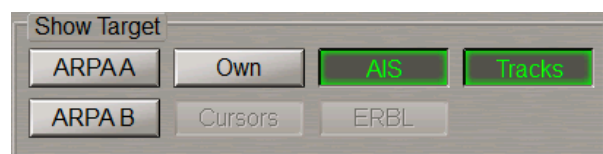
Name	A2	273002414	671641000
Alias		Fellow	
CPA (NM)	1.90	3.24	1.95
TCPA (min)	XXXX	-79.8	XXXX

On the Chart panel, the target will be displayed with the entered “Alias” identifier regardless of settings made in the **AIS Target identification** group. This identifier is displayed on the Chart panel only and is not transmitted in messages.

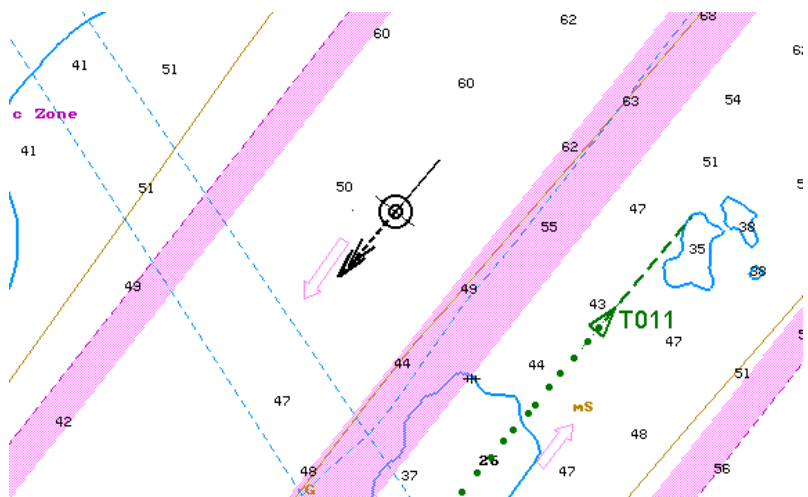


10.3.1.3. Display of Target Tracks

Turn on, as required, the display of target tracks. To do this, press the **Tracks** button in the **Show Target** group.



Target tracks will be displayed on the Chart panel.

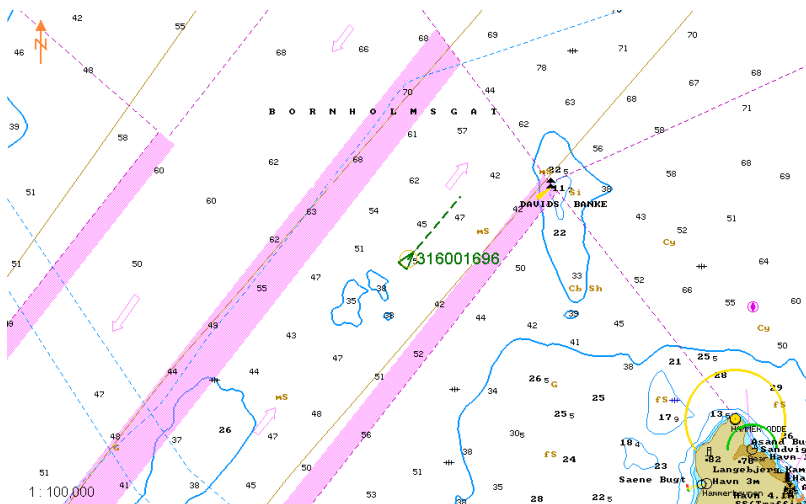


10.3.1.4. Prompt Target Search

Press the button with the target identifier in the **Name** line of the Targets Table, e.g., 316001696.

Name	273002414	316001696	671641000
------	-----------	-----------	-----------

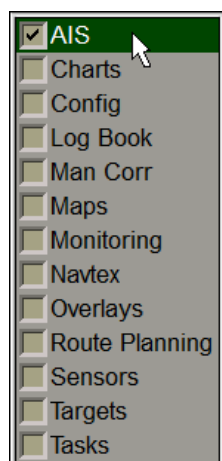
The selected target will be displayed in the centre of the Chart panel highlighted with a flashing circle.



10.3.2. Setting of Voyage Data

ATTENTION! Setting of voyage data is available only on the station with the MASTER status.

Open the **AIS** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Static Data:		Restore
Name:	KOLA	
Call Sign:	UCKB	
MMSI:	273130500	
IMO Number:	008013053	

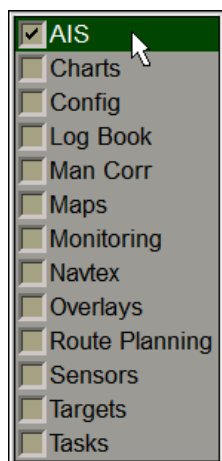
As other devices are connected to the AIS transponder in addition to the MFD (e.g., MKD or pilot software), the own ship name and call sign in the transponder can be changed by these users. The operator is notified about these changes by the **Check AIS Ship Name** and **Check AIS Call Sign** cautions which will remain active until the own ship data is restored in the transponder. To restore the own ship data settings in the transponder as per the settings in the System Configuration utility, press the **Restore** button.

The setting of own ship parameters transmitted in the AIS system is completed.

10.3.3. Transmission of Messages in AIS System

Turn on the display of AIS targets (see paragraph *Handling AIS Targets*).

Open the **AIS** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the left-hand part of the **AIS** panel to switch to the **Messaging** page.

Voyage and Static Data		Messaging		IMO special messages		Interrogation		Long Range		Channel Management	
Send message											
Create Message											
N	Sent to	Date/Time (UTC)	Type	Status							
3	316001696	20-04-2006 10:04:49	Safety Text	Waiting							
2	273002413	20-04-2006 10:04:27	Safety Text	Delivered							
1	ALL	20-04-2006 10:01:43	Safety Text	Sent							

Receive message	
From:	273002413
To:	All stations
Type:	Safety text
THE RACON ON LIGHTBUOY TANGO-1 IN PSN 5747 2 N 01046.1 E IS INOPERATIVE	
☒ All message notification ☒ Show new message	
Reply Request Route Show Sender Previous Next	

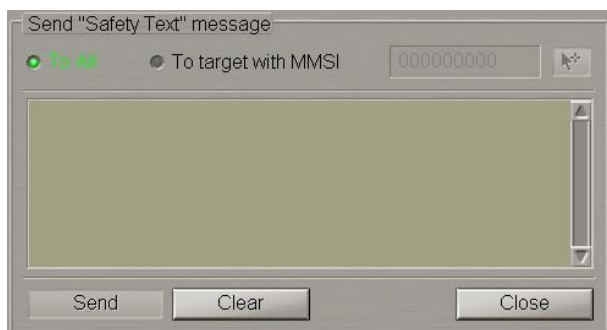
To prepare and transmit a text message, press the **Create Message** button on the **Messaging** page in the **Send message** group and select **Safety Text**.

NOTE: If the class B AIS is selected during the installation, the capability to send messages with the "Safety Text" and/or "Normal Text" status is set in the System Configuration utility. (See *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Installation Guide, Chapter 2, section MFD Configuration, paragraph Sensors Settings*).



Press the left trackball button.

The **Send Message** window will open up, specifying the type of the selected message.



Set the identifier of the addressee of the message to be sent. To transmit a message to all the ships in the radio coverage area, select the **To All** checkbox.



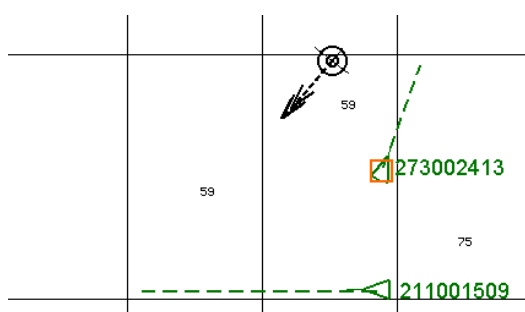
To transmit a message to a certain ship, use one of the three procedures:

1. Select the **To target with MMSI** checkbox and enter the MMSI of the addressee in the input line which will be activated.



2. Select the **To target with MMSI** checkbox and press the  button to the right of the input line.

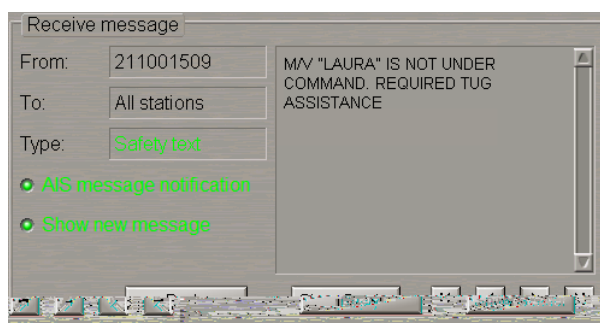
Position the acquisition marker, which will appear on the Chart panel, on the target, which is the addressee of the message.



Press the left trackball button. The input line will display the MMSI of the specified target.



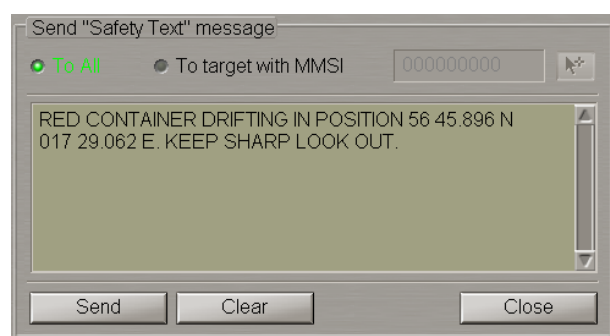
3. To send a reply message, press the **Reply** button in the **Receive Message** group.



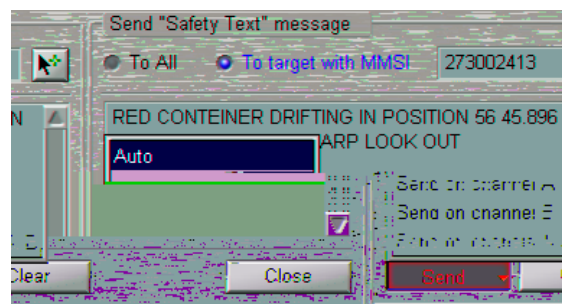
With the use of the **Reply** function, the **Send message** window opens up for a message, which has the same type as the message received from an AIS object, "Safety Text" in this case. The input line will automatically display MMSI of the AIS object, which the message has been received from.



Use the text input window in the **Send Message** group to enter the text of the message which should be sent.



Press the **Send** button.



Select the message despatch channel (**Auto** is generally recommended). The compiled message will be sent.

Press the **Close** button.

Check the sent message status in the table in the **Send message** group. The message status may be one of the following:

- **Waiting** – the message has been transmitted by the MFD, but no confirmation of its relay has been received from the transponder;
- **Sent** – the message has been sent by the transponder, but no confirmation of its delivery to the addressee has been received or confirmation is not expected if the message was sent to all the AIS objects (“All” identifier);
- **Delivered** – the message has been delivered to the addressee;
- **Failed** – the message, for some reasons, cannot be sent by the transponder.

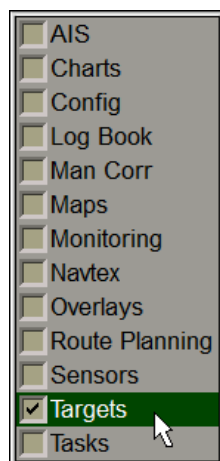
The text of sent “Safety Text” messages, as well as rected

10.4. Handling Targets

10.4.1. Setting of Safety Parameters in Tracking of Targets

ATTENTION! In the ECDIS task with the radar overlay (no Navi-Radar 4000 installed and running) CPA/TCPA alarm cannot be generated. If the RADAR task installed and running, immediately after the RADAR task shuts down, all the CPA/TCPA alerts get rectified and due to the loss of the collision avoidance function provided by the Radar task. For external ARPA targets (ARPA_A, ARPA_B) CPA/TCPA alarms cannot be generated. No danger status is displayed for such targets. (For more information see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 4, section Alert Management Principles, paragraph General*).

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

Name	40	41	42	43	44	37	47	48	49	50	51	52	53	54
Alias														
CPA (NM)	2.88	5.87	7.32	6.81	7.74	5.81	4.09	4.53	5.86	8.23	1.25	6.76	8.07	5.62
TCPA (min)	-9.7	-8.4	-6.9	-14.5	-9.3	-21.2	-21.2	-21.2	10.4	12.8	21.8	-16.4	-16.6	-19.2
SOG (kn)	45.3	43.5	29.4	26.8	33.1	11.3	15.0	13.1	22.4	12.3	16.8	19.5	15.8	12.8
COG (°)	145°	295.9°	65.7°	287.1°	281.1°	30.0°	5.9°	36.4°	84.2°	170.7°	81.6°	348.7°	358.2°	29.6°
Range (NM)	8.94	8.33	8.43	9.10	9.05	8.79	8.67	8.57	7.63	8.46	8.44	9.59	10.11	8.56
Bearing (°)	359.3°	350.7°	0.3°	351.7°	352.5°	352.7°	45.2°	5.8°	203.4°	236.6°	257.5°	317.5°	318.4°	352.4°
Bow X (NM)	7.7	-5.9	-20.9	-6.9	-8.1	55.4	-10.5	112.1	10.6	8.2	-2.6	11.1	16.8	49.3
TBow X (min)	-17.9	-9.0	25.4	-16.5	-13.8	-197.9	5.6	-346.8	-8.4	15.6	27.7	-37.6	-56.9	-164.8

In the top of the **Targets** panel which will open up, turn on the display of targets from the required sources by pressing the appropriate buttons: **ARPA A(B)**, **Own** or **AIS**.



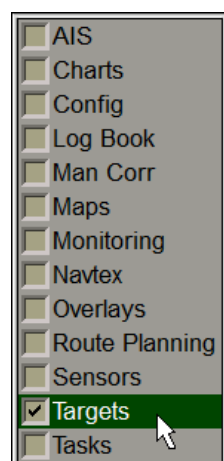
The **Alarm** group contains windows for entering CPA/TCPA values.

The **CPA/TCPA Tgt ID** alarm generation is turned on when both CPA and TCPA values of the target are smaller than the set values:

- **CPA** – to enter the minimum closest point of approach distance;
- **TCPA** – to enter the minimum time to the closest point of approach.

10.4.2. Handling Target Table

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

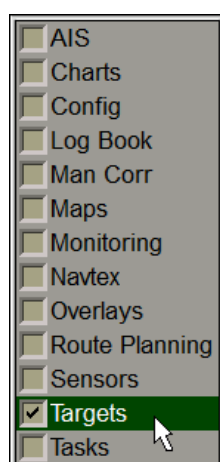
Name	71	74	75	77	78	79	80	91	92	93	95	100	138	182
Alias	Fellow													
CPA (nm)	3.08	2.39	2.01	1.04	0.74	0.07	1.94	3.29	3.70	3.67	1.49	1.33	3.45	0.64
TCPA (min)	-2.5	11.5	1.1	-3.1	-2.3	1.3	3.7	0.8	0.5	-0.3	-1.9	1.4	1.3	8.9
COG (°)	223.6°	335.2°	354.1°	202.5°	219.0°	340.8°	222.2°	268.9°	264.1°	244.8°	292.1°	23.1°	336.6°	336.7°
SOG (kn)	91.7	8.5	32.6	13.6	9.5	9.3	5.3	37.1	42.2	51.0	39.5	22.9	37.5	19.1
Range (nm)	5.18	3.82	2.11	1.46	1.00	0.34	2.12	3.35	3.72	3.68	2.16	1.41	3.57	3.75
Bearing (°)	264.4°	158.2°	85.4°	185.3°	197.6°	135.9°	140.1°	10.3°	2.0°	334.9°	331.3°	107.4°	68.9°	144.8°
Bow X (nm)	-4.6	-4.8	2.2	1.6	1.8	-0.1	7.0	-213.1	-44.4	-10.2	4.9	1.3	4.3	-0.9
TBow X (min)	-0.5	27.6	-0.3	-7.0	-7.7	1.7	-25.0	272.3	51.4	9.2	-7.7	1.2	-2.2	10.5

The Target Table contains columns (ARPA and AIS targets) and rows with the targets information with a capability to sort by the relevant parameter:

- **Name** – target identifier;
- **Alias** – target identifier (up to 8 characters) assigned by the operator (the target is displayed on the Chart panel with this identifier regardless of settings in the **AIS Target Identification** group);
- **CPA** – closest point of approach distance;
- **TCPA** – time to the closest point of approach;
- **COG** – target course over the ground;
- **SOG** – target speed over the ground;
- **Range** – distance to the target;
- **Bearing** – bearing to the target;
- **Bow X** – distance to the point where the target crosses the own ship course;
- **TBow X** – time to the point where the target crosses the own ship course.

10.4.3. Prompt Target Search

Open the **Targets** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



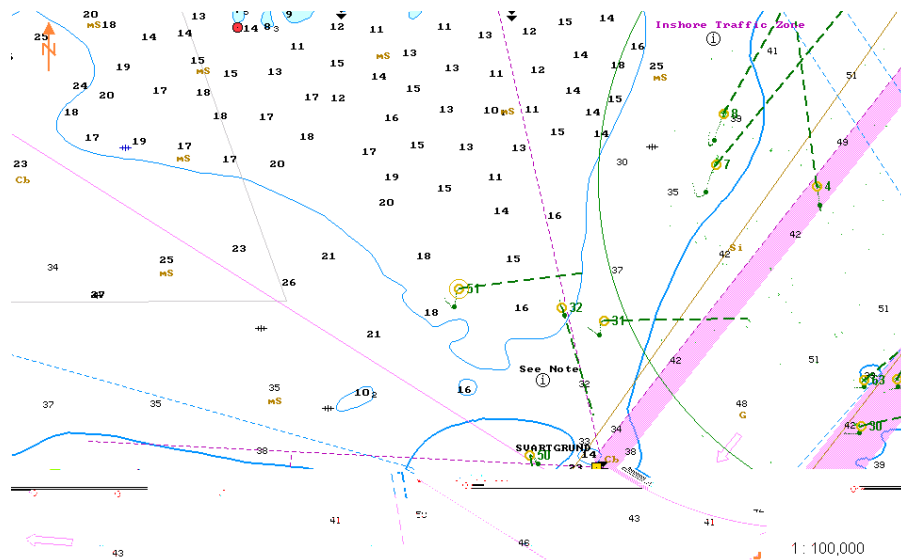
Use the tab in the top part of the **Targets** panel which will open up, to switch to the **Target Table** page:

Target Table		AIS color tracks	
Show Target		AIS Target identification	
ARPA A	ARPA B	By Name	Alarm
ARPA B	Cursors	Max. range	CPA 2.00 NM
	ERBL	48 nm	TCPA 12.0 min
Name	40	41	42
Alias			
CPA (NM)	2.88	5.87	7.32
TCPA (min)	-9.7	-8.4	-6.9
COG (°)	14.5°	295.9°	65.7°
SOG (kn)	45.3	43.5	29.4
Range (NM)	8.94	8.33	8.43
Bearing (°)	359.3°	350.7°	0.3°
Bow X (NM)	7.7	-5.9	-20.9
TBow X (min)	-17.9	-9.0	25.4

Turn on the display of targets (see section *AIS*, paragraph *Handling AIS Targets* and section *ARPA*, paragraph *Target Display*).

Press the button with the target identifier in the **Name** line of the Targets Table, e.g., “51”.

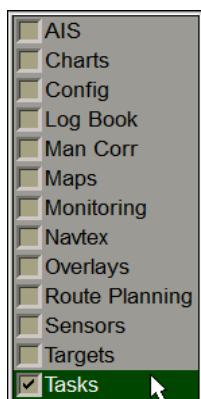
The selected target will be displayed in the centre of the Chart panel highlighted with a flashing circle.



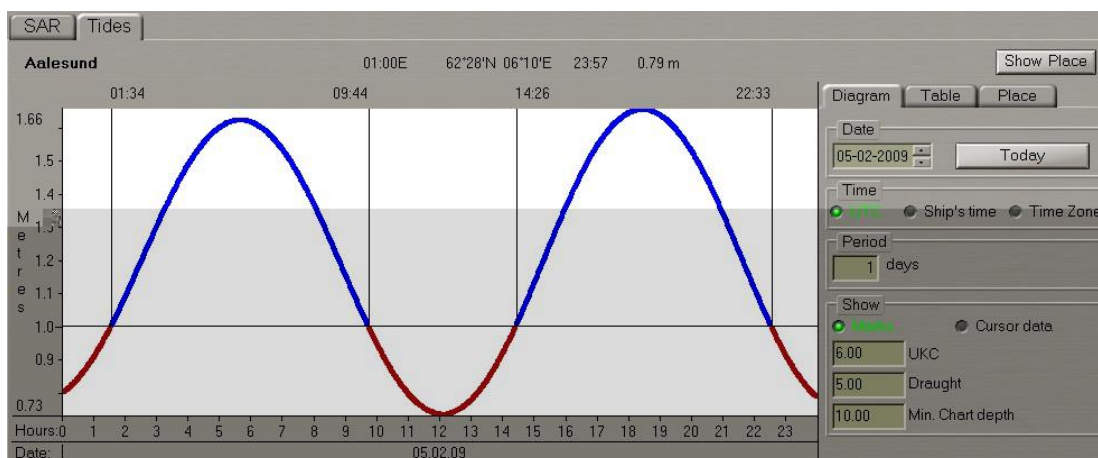
11.5. Obtaining Information on Tides and Currents

11.5.1. Obtaining Information on the Change of Tidal Heights

Open the **Tasks** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



Use the tab in the top part of the **Tasks** panel to switch to the **Tides** page.



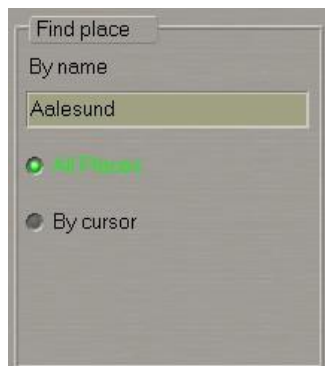
Use the tab in the top right part of the **Tides** page which will open up, to switch to the **Place** tab.

A screenshot of the 'Place' tab in the 'Tides' page. The top bar shows 'SAR' and 'Tides' tabs. Below the tabs, the location 'Aalesund' is displayed with coordinates '01:00E 62°28'N 06°10'E'. The main area shows a list of places with columns 'Place', 'Coordinates', and 'Time Zone'. The right side of the panel has a 'Set Active' button and a 'Show Place' button. Below the buttons, there are tabs for 'Diagram', 'Table', and 'Place'. The 'Place' tab is active. Below the tabs, there is a 'Find place' section with a 'By name' input field containing 'Aalesund'. There are also radio buttons for 'All Places' and 'By cursor'.

Place	Coordinates	Time Zone
Aalesund	62°28'N 06°10'E	01:00E
Abaiang Atoll	01°49'N 173°02'E	12:00E
Abashiri Ko	44°01'N 144°17'E	09:00E
Abbot Point	19°53'S 148°05'E	10:00E
Abermama	00°29'N 173°52'E	12:00E
Aber Benoit	48°35'N 04°37'W	01:00E
Aberdaron	52°48'N 04°43'W	00:00
Aberdeen	46°58'N 123°51'W	08:00W
ABERDEEN	57°09'N 02°05'W	00:00
Aberdovey	52°32'N 04°03'W	00:00
Aberporth	52°08'N 04°33'W	00:00
Aberystwyth	52°24'N 04°05'W	00:00
Abidjan Entrance	05°15'N 03°60'W	00:00
Ahlaton Lake	70°50'S 68°25'W	03:00W

In the right-hand part of the **Place** tab which will open up, select the reference point by using one of the following procedures:

- Enter the reference point name in the **By name** input line and press <ENTER>;
- Select the **All Places** checkbox.

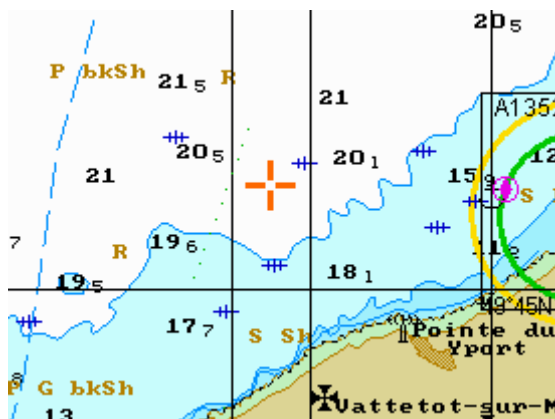


In the list in the left-hand part of the page which will open up, select the reference point and double click the left trackball button.

- Select the **By cursor** checkbox; enter the radius of the area coverage circle in the **Distance** input line which will open up.



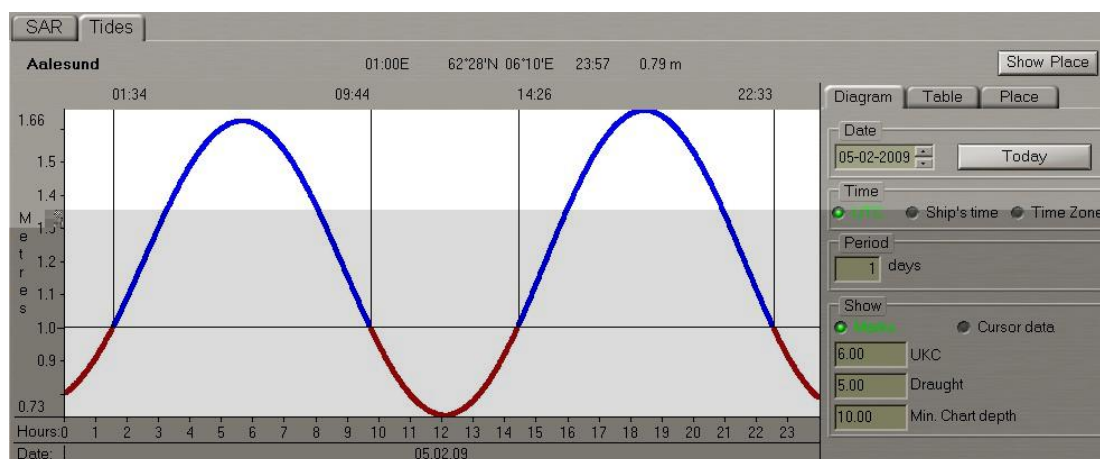
Press the **Find** button. Position the graphic cursors which will appear, on the point within the Chart panel which will be the centre of the area coverage circle.



Press the left trackball button. In the list of reference points which fall within the coverage area in the left-hand part of the page, select the reference point and double click the left trackball button.

11.5.1.1. Information Displayed Graphically

After the selection of the reference point, the **Diagram** tab will open up, displaying the tidal curve for the selected reference point.



In the **Date** line of the **Diagram** tab, select the date starting from which the tidal heights will be calculated. Use the **Period** input line to enter the period which tidal heights are required to be calculated for.

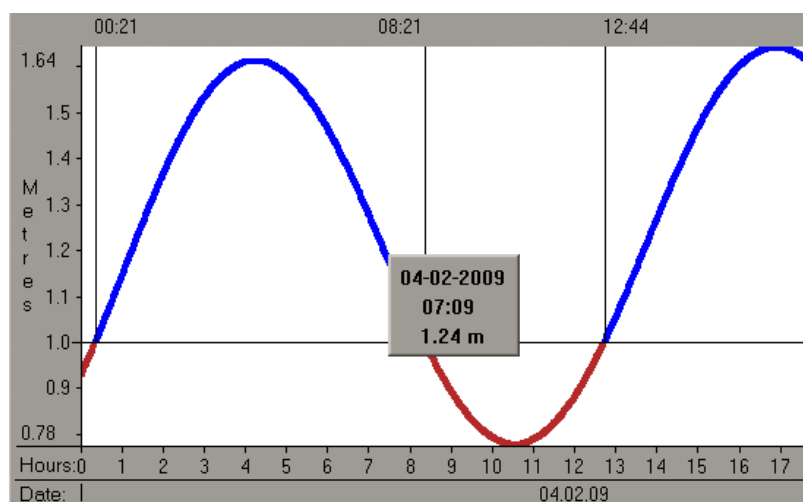
The following parameters can be set:

- **UKC** – under keel clearance;
- **Draught** – ship's draught;
- **Min. Chart depth** – the minimum depth on the chart.

Select the **Marks** checkbox to show the time marks on the diagram.

On the graph, blue is used for showing the time periods when the actual depth is safe, and red for showing the time intervals with dangerous depths.

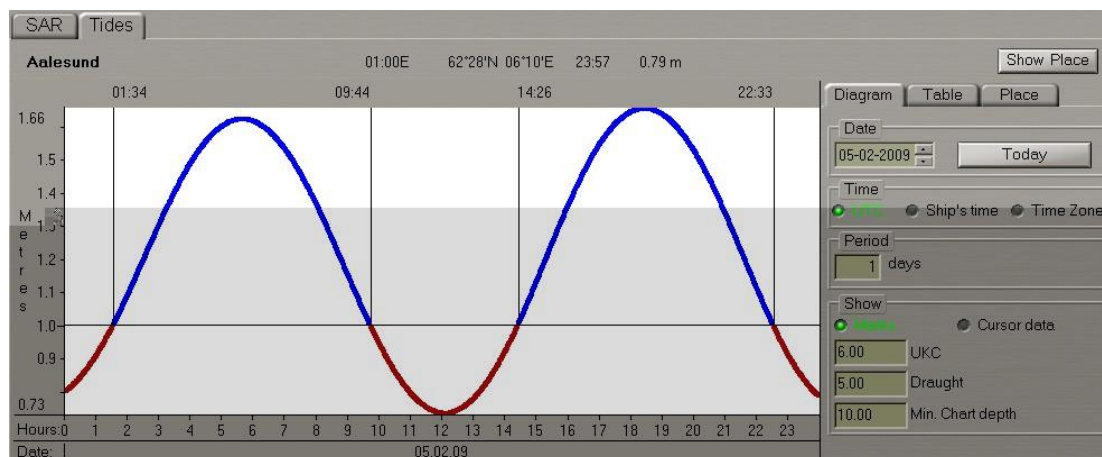
Select the **Cursor Data** checkbox. Position the cursor on the tidal curve and by moving the cursor with the trackball, set the necessary time value in the cursor information window.



Read the tidal height value in the cursor information window.

11.5.1.2. Information Displayed in Tabular Form

After the selection of the reference point, the **Diagram** tab will open up displaying the tidal curve for the selected reference point.



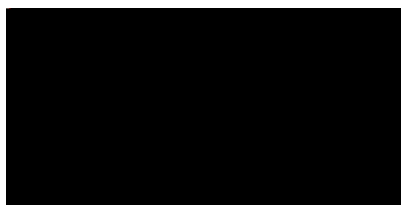
Use the tab in the top right part of the page to switch to the **Table** tab.

UTC TIME	Heights of water in Metres for high and low waters
04:15	1.62
10:33	0.78
16:55	1.64
23:23	0.78

In the **Date** line of the **Table** tab, select the date starting from which the tidal heights will be calculated. Use the **Period** input line to enter the period which tidal heights are required to be calculated for.

To calculate tidal tables, use one of the following procedures:

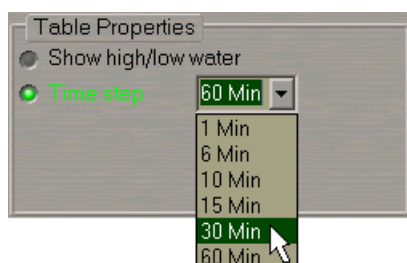
- In the Table Properties group, select the Show high/low water checkbox.



The left-hand part of the page will display a table of high and low water.

Heights of water in Metres for high and low waters	
Aalesund	
62°28'N 06°10'E	
UTC TIME	
4 February 2009	
04:15	1.62
10:33	0.78
16:55	1.64
23:23	0.78

- In the **Table Properties** group, select the **Time step** checkbox; press the  button in the activated line for the input of the tidal height calculations discretion.



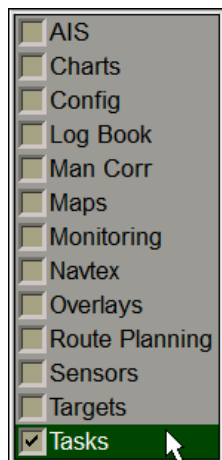
In the list which will open up, select the necessary tidal height calculation discretion and press the left trackball button. The left-hand part of the page will display the tidal height table calculated with the set discretion.

Heights of water in Metres for regular intervals	
Aalesund	
62°28'N 06°10'E	
UTC TIME	
4 February 2009	
00:00 0.93	04:00 1.61
00:30 1.03	04:30 1.61
01:00 1.15	05:00 1.59
01:30 1.26	05:30 1.54
02:00 1.37	06:00 1.46
02:30 1.46	06:30 1.38
03:00 1.54	07:00 1.28
03:30 1.59	07:30 1.17
08:00 1.07	11:00 0.79
08:30 0.97	11:30 0.82
09:00 0.89	12:00 0.88
09:30 0.83	12:30 0.96
10:00 0.79	13:00 1.05
10:30 0.78	13:30 1.16
11:00 0.79	14:00 1.27
11:30 0.82	14:30 1.37
12:00 0.88	15:00 1.47
12:30 0.96	15:30 1.55
13:00 1.05	16:00 1.60
13:30 1.16	16:30 1.64
14:00 1.27	17:00 1.64
14:30 1.37	17:30 1.63
15:00 1.47	18:00 1.59
15:30 1.55	18:30 1.52
16:00 1.60	19:00 1.44
16:30 1.64	19:30 1.34
17:00 1.64	20:00 1.24
17:30 1.63	20:30 1.13
18:00 1.59	21:00 1.03
18:30 1.52	21:30 0.94
19:00 1.44	22:00 0.87
19:30 1.34	22:30 0.81
20:00 1.24	23:00 0.78
20:30 1.13	23:30 0.78

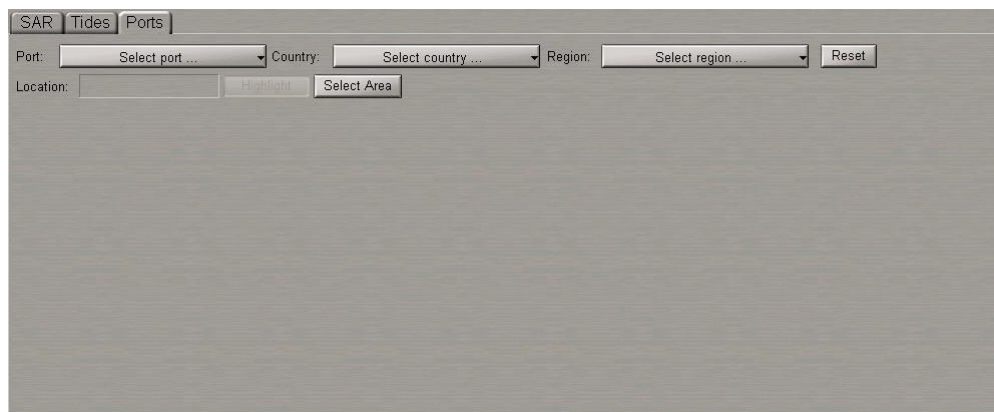
Take the necessary readings in the calculated tables.

11.8. Obtaining Information on Ports

Open the **Tasks** panel by selecting the appropriate line of the **Tasks List** menu in the tabs zone of the ECDIS task screen.



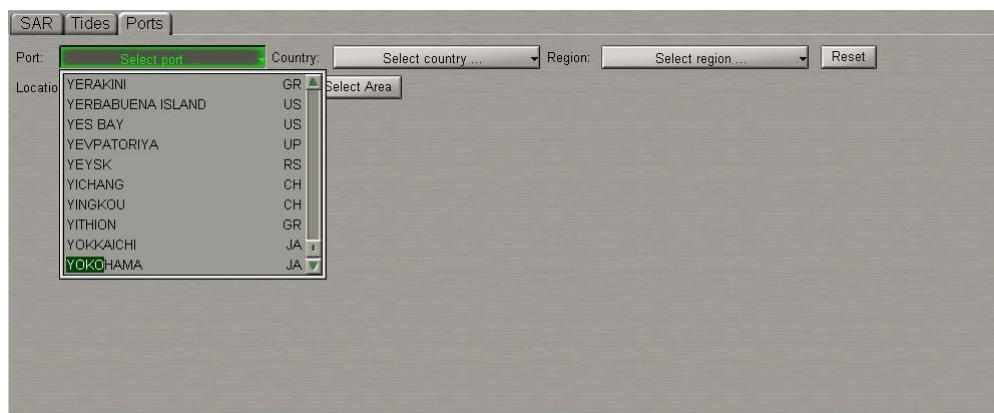
Use the tab in the top part of the **Tasks** panel to switch to the **Ports** page:



To obtain information on the required port, use the procedure described below.

11.8.1. Obtaining Information by Port Name

Press the **Port** button. Type the first letters of the port name or select the port from list with the cursor.



Press left trackball button or <ENTER>. Information on the selected port will be displayed on the page.

The screenshot shows the 'Ports' tab selected. The 'Port' field contains 'YOKOHAMA', 'Country' is 'Japan', and 'Region' is 'JAPAN'. A 'Reset' button is visible. Below these fields are 'Location' (35°27' N 139°35' E) and 'Highlight'/'Select Area' buttons. The 'General info' tab is active, displaying details for YOKOHAMA:

- Harbour:** Type: coastal breakwater, Size: large, Shelter: good.
- Entrance restrictions:** Depths: Main channel: 12.4 m - 13.7, Cargo piers: 15.5 m - 16.7, Oil terminal: 18.5 m - 19.8, Anchorage: 17.0 m - 18.2. Mean level of tide: 1.2 m, Max. size vessel: over 152 m in length.
- Formalities:** ECA message, No port of entry, Load/offload, Anchor, Med moor, Beach moor, Ice moor.
- Quarantine:** Quarantine, Salvage.
- Pilotage:** Compulsory, Recommended, Local assistance.

Read the necessary information (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 7, section Ports*). For obtaining information on other ports, press the **Reset** button and repeat the procedure.

11.8.2. Obtaining Information by Region Name

Press the **Region** button. Type the first letters of the required region or select it from list with the cursor.

The screenshot shows the 'Region' dropdown menu open, displaying a list of regions. The 'Region' field currently shows 'Select region ...'. The list includes:

- SAUDI ARABIA
- SAUDI ARABIA CONTINUED
- SCOTLAND EAST COAST
- SCOTLAND NORTH COAST
- SCOTLAND WEST COAST
- SENEGAL
- SERAM
- SEYCHELLES
- SHETLAND ISLANDS
- SICILIA

Press left trackball button or <ENTER>. Press the **Port** button. In this case, the list displays ports belonging to the selected region. Type the first letters of the port name or select the port from the list with the cursor.

The screenshot shows the 'Port' dropdown menu open, displaying a list of ports for the selected region 'SICILIA'. The 'Port' field currently shows 'Select port ...'. The list includes:

- AUGUSTA
- CATANIA
- FAVIGNANA
- GELA
- GIOIA TAURO
- LICATA
- MARSALA
- MAZARA DEL VALLO
- MESSINA
- MILAZZO

Press the left trackball button or <ENTER>. Information on the selected port will be displayed on the page.

The screenshot shows the 'Ports' tab selected. The 'Port' field contains 'MESSINA', 'Country' is 'Italy', and 'Region' is 'SICILIA'. The 'Location' is '38°12' N 015°34' E'. Below this, there are two tabs: 'General info' and 'Facilities & services'. The 'General info' tab is active, showing details about the harbour (coastal natural, medium size, good shelter), pilotage (compulsory, available), entrance restrictions (depths, cargo piers, oil terminal, anchorage), formalities (ETA messages, pre-arrival, load/offload, moorings), quarantine (clear certificates, phosphate), and tugs (salvage, anchor).

Read the necessary information (see document *Multifunctional Display. Navi-Sailor 4000 ECDIS (v. 3.02.350). Functional Description, Chapter 7, section Ports*). For obtaining information on other ports, press the **Reset** button and repeat the procedure.

11.8.3. Obtaining Information by Country Name

Press the **Country** button. Type the first letters of the required country or select it from the list with the cursor.

The screenshot shows the 'Ports' tab. The 'Country' field is highlighted, and a list of countries is displayed. The list includes: French Southern and Antarctic Lands, Gabon, Gambia, The, Gaza Strip, Georgia, Germany, Ghana, Gibraltar, Glorious Islands, and Greece. 'Greece' is highlighted at the bottom of the list.

Press the left trackball button or <ENTER>. Press the **Port** button. In this case, the list displays ports belonging to the selected country. Type the first letters of the port name or select the port from list with the cursor.

The screenshot shows the 'Ports' tab. The 'Country' field is 'Greece', and the 'Region' is 'Select region ...'. A list of ports is displayed, including: ORMOS VATHI, OROPOS, PACHI OIL TERMINAL, PALIOURION, PARALION ASTROS, PAROS, PATRAI, PERAMA, PILION, and PIRAEVS. 'PIRAEVS' is highlighted at the bottom of the list.